

A-DACS: AI-POWERED DIGITAL CLEARANCE SYSTEM

A PROJECT REPORT Submitted by

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TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	LIST OF FIGURES	viii
	LIST OF TABLES	ix
	ABSTRACT	1
1	INTRODUCTION	2
	1.1 Background	2
	1.2 Problem Statement	3
	1.3 Objectives	3
	1.3.1 Primary Objectives	4
	1.3.2 Secondary Objectives	4
	1.4 Scope of the Project	5
	1.5 Project Contribution	5
2	LITERATURE REVIEW	6
	2.1 Review of Recent Articles	7
	2.2 Comparative Analysis of Existing Approaches	8
	2.3 Summary of Literature Survey	10
3	SYSTEM ANALYSIS	11
	3.1 Existing System	11
	3.1.1 Overview of Current Clearance Systems	11
	3.1.2 Disadvantages of the Existing System	12
	3.2 Proposed System	12
	3.3 Feasibility Study	13
4	SYSTEM REQUIREMENTS	14
	4.1 Hardware Requirements	14

	4.2 Software Requirements	15
	4.3 Functional Requirements	16
	4.4 Non-Functional Requirements	17
5	SYSTEM DESIGN	18
	5.1 Overview of A-DACS	18
	5.2 System Architecture	19
	5.3 Advantages of the Proposed System	20
6	MODULE DESCRIPTION	21
	6.1 Methodology Overview	21
	6.2 Modules	21
	6.2.1 Module 1: User Management & Authentication	22
	6.2.2 Module 2: Payment Processing & Fraud Detection	24
	6.2.3 Module 3: Certificate & Communication Management	25
7	IMPLEMENTATION	27
	7.1 System Setup	27
	7.2 Project Code Structure	37
	7.3 Workflow Execution	37
	7.4 Integration Overview	38
	7.5 Key Implementation Challenges and Solutions	39
8	RESULTS AND SNAPSHOTS	41
	8.1 User Interface Snapshots	41
	8.1.1 Login and Registration Screens	42
	8.1.2 Student and Staff Dashboard	43
	8.1.3 Payment and Receipt Upload	45
	8.1.4 OCR Extraction Results	46
	8.1.5 Admin Verification Panel	47
	8.1.6 Certificate Generation	48

9	TESTING	49
	9.1 Unit Testing	49
	9.2 Integration Testing	49
	9.3 User Acceptance Testing	50
	9.4 Performance Testing	51
10	CONCLUSION AND FUTURE WORK	52
	10.1 Conclusion	52
	10.2 Future Enhancements	52
	REFERENCES	53
	CERTIFICATES	55

LIST OF TABLES

CHAPTER NO	TITLE	PAGE NO
2	2.2 Comparative Analysis Of Existing Approaches	9
4	4.1 Hardware Requirements	17
	4.2 Software Requirements	17
5	5.2 Technology Stack	22
6	6.2.2 Regex Pattern Library	29
7	7.1 System Setup - Prerequisites	35
	7.2.1 Project Code – Main Components	37
	7.2.2 Services Layer	38
	7.4 Firebase Collections	40
9	9.1 Unit Test Coverage Areas	46
	9.2 Integration Test Results	46
	9.3 Security Test Results	47
	9.4 Performance Results	48

LIST OF FIGURES

CHAPTER NO	TITLE	PAGE NO
5	5.2 System Architecture	22
6	6.2 Core Modules Classification	26
7	7.4 OCR Workflow	41
8	8.1.1 Splash and Onboarding Screens	42
	8.1.2 Admin Dashboard Screen	43
	8.1.3 Register and Login Screens	44
	8.1.4 Student and Staff Dashboard Screens	45
	8.1.5 Payment and Receipt Upload Screens	45
	8.1.6 OCR Details Extraction Screen	46
	8.1.7 Admin Approval Screen	46
	8.1.8 No-due Certificate Generation Screen	47

ABSTRACT

Educational institutions globally face significant challenges in managing student clearance processes, with traditional paper-based systems leading to severe administrative inefficiencies, financial discrepancies, and widespread certificate forgery. Manual clearance systems result in substantial revenue leakage through unverified payments and remain highly susceptible to document tampering, while current digital solutions lack integrated fraud detection mechanisms, leaving institutions vulnerable to manipulated payment receipts and counterfeit clearance certificates. Parents remain disconnected from financial workflows, receiving no real-time updates about their ward's fee status or impending deadlines, creating communication gaps that lead to missed payments and unnecessary penalties. The absence of automated verification creates administrative bottlenecks during peak clearance periods, causing student dissatisfaction and institutional compliance issues. This research introduces A-DACS (AI-powered Digital Clearance System), a comprehensive role-based application that revolutionizes educational clearance through artificial intelligence and cloud technologies. The system combines on-device OCR text extraction using Google ML Kit with a sophisticated probability scoring algorithm to detect payment receipt tampering through cross-referencing extracted and user-submitted data, flagging discrepancies for administrative review. The platform implements cryptographically unique UUID-based certificates with QR verification, enabling public validation against live database records, ensuring any forgery attempt is immediately detectable. A serverless cloud architecture utilizing Firebase Cloud Functions with automated multichannel notifications through Twilio SMS, Email, and Push notifications ensures 24/7 parent-student communication, while smart scheduling algorithms trigger reminders at 7, 3, and 1 days before deadlines. The dynamic fee engine allows administrators to configure complex fee structures based on batch, department, and quota combinations, while the integrated wallet system manages overpayments seamlessly across semesters. Unlike existing fee management systems that merely digitize records, A-DACS provides comprehensive algorithmic fraud detection, tamper-proof certification with cryptographic security, and automated parent engagement through Twilio-integrated SMS workflows compliant with Indian regulations. The system's real-time verification mechanisms, combined with departmental dashboards and comprehensive audit trails, establish a new benchmark for secure, transparent educational clearance processes adaptable for institutions worldwide.

CHAPTER 1

INTRODUCTION

This chapter provides a comprehensive overview of the A-DACS (AI-powered Digital Clearance System) project, establishing the foundation for understanding the research. It begins by exploring the background and context of educational clearance processes, followed by a detailed articulation of the problems prevalent in current manual systems. The chapter then outlines the primary and secondary objectives that guide the development, defines the functional and technical scope of the project, and concludes with the unique contributions this research makes to the field of educational technology.

1.1 Background

The digitization of administrative processes in educational institutions has become increasingly critical in an era where technology permeates every aspect of daily life. Student clearance systems, which encompass fee verification, no-due certifications, and final document issuance, represent one of the most crucial yet traditionally neglected areas of institutional management. These processes serve as the final checkpoint before students graduate, transfer, or proceed to higher education, making their accuracy and security paramount. Educational institutions have historically relied on manual, paper-based clearance systems that require students to visit multiple departments, collect signatures, submit physical payment receipts, and await manual verification from administrative staff.

Recent technological advancements in artificial intelligence, cloud computing, and mobile application development have opened new possibilities for transforming educational administration. Machine learning algorithms can now analyze payment receipts with remarkable accuracy, while cloud platforms enable real-time data synchronization and automated communication. Mobile applications provide ubiquitous access to institutional services, allowing students and parents to monitor financial status and receive instant notifications about important deadlines. These technological capabilities create an opportunity to reimagine the entire clearance process as a seamless, secure, and transparent digital workflow. A-DACS (AI-powered Digital Clearance System) emerges from this technological convergence, specifically designed to address the unique challenges of educational clearance processes. By leveraging on-device OCR technology, probability scoring algorithms, and serverless cloud architecture, the system aims to eliminate paperwork, prevent fraud, and enhance communication between institutions, students, and parents.

1.2 Problem Statement

Educational institutions today face a multifaceted crisis in their student clearance processes that manifests in several critical areas. The manual verification of fee payments and clearance certificates has proven to be not only time-consuming but also highly vulnerable to manipulation and error. Students frequently experience delays in receiving their no-due certificates, which

directly impacts their ability to pursue higher education or secure employment opportunities immediately after graduation.

The financial integrity of institutions remains at significant risk due to the absence of robust verification mechanisms for payment receipts. Students can easily manipulate digital screenshots of payment confirmations using basic photo editing software, altering transaction amounts, dates, or reference numbers to show payments that were never made or were made for lesser amounts. Current manual verification processes, which rely on visual inspection by overworked administrative staff, cannot consistently detect such manipulations, leading to substantial revenue leakage and potential legal complications.

Furthermore, the lack of transparent communication channels between institutions and parents creates significant operational challenges. Parents, who are often the primary financial supporters of students' education, receive no real-time updates about fee deadlines, payment confirmations, or outstanding dues. This information gap results in missed payments, late fee penalties, and unnecessary friction between institutions and families. When payment issues arise, institutions struggle to reach parents promptly, exacerbating conflicts and delaying resolutions.

Administrative staff face overwhelming workloads during peak clearance periods, manually verifying hundreds of payment receipts, cross-referencing multiple departmental records, and generating certificates through time-consuming manual processes. This administrative burden not only delays service delivery but also increases the likelihood of human error, with genuine payments occasionally going unrecorded or certificates containing incorrect information.

A-DACS addresses these interconnected problems by creating an integrated digital ecosystem that automates verification, secures documentation, and ensures transparent communication throughout the clearance process. The system's AI-powered receipt analysis detects manipulated submissions automatically, while cryptographic certificate generation prevents forgery. Automated notifications keep parents informed at every stage, and role-based dashboards enable administrative staff to monitor progress efficiently.

1.3 Objectives

This section defines the clear, measurable goals that guide the entire research and development process of A-DACS. The objectives are structured hierarchically to distinguish between the fundamental goals that form the core of the system and the additional functionalities that enhance its value proposition.

1.3.1 Primary Objectives

1. **To develop an AI-powered receipt verification system** that automatically extracts transaction details from payment screenshots using OCR technology and validates their authenticity through cross-referencing with expected fee amounts and pattern recognition algorithms, thereby eliminating manual verification and detecting fraudulent submissions.

2. **To implement a cryptographically secure certificate generation mechanism** that creates tamper-proof no-due certificates with unique UUID identifiers and QR codes, enabling instant public verification against live database records and preventing document forgery.
3. **To establish an automated multichannel notification framework** that delivers real-time updates to students and parents through SMS, email, and push notifications regarding fee deadlines, payment confirmations, verification status, and certificate availability, ensuring continuous communication without administrative intervention.

1.3.2 Secondary Objectives

1. **To create a dynamic fee management engine** that enables administrators to configure complex fee structures based on multiple parameters, including academic batch, department, student quota, and category, while automatically calculating individual student dues and tracking payment history across semesters.
2. **To design role-based access control interfaces** that provide customized dashboards for students, departmental staff, and administrators, ensuring each user group has appropriate visibility and functionality aligned with their responsibilities in the clearance process.
3. **To implement a comprehensive audit trail system** that maintains immutable records of all payment verifications, certificate generations, and administrative actions, facilitating financial compliance and enabling retrospective analysis of institutional clearance operations.
4. **To integrate a digital wallet functionality** that automatically manages student overpayments by storing excess amounts for future semester adjustments, eliminating the need for physical refunds and ensuring accurate fee accounting.

1.4 Scope of the Project

- User management with manual registration requiring administrative approval for all accounts and role-based access control for Students, Staff, and Administrators. Student payment processing supporting UPI intent, QR code, and Demand Draft with automatic OCR extraction of transaction details, including dates, amounts, and transaction IDs
- AI-powered fraud detection using Google ML Kit with keyword validation, probability scoring, and tampering flags for manually edited fields. Cryptographic certificate generation with UUID-based unique identifiers and QR codes enabling instant public verification against live database records
- Staff departmental dashboards showing real-time student payment status with colour-coded indicators for PAID, OVERDUE, and PENDING
- Administrative control panel for user approval, dynamic fee configuration, payment audit with tampering alerts, and bulk communication triggers. Automated multichannel communication through Twilio SMS, email, and Firebase Cloud Messaging.

1.5 Project Contribution

- Introduces AI-powered fraud detection through on-device OCR and probability scoring algorithms that actively analyze receipts and flag manipulated documents, unlike existing systems that merely store them
- Implements cryptographic certificate security with UUID-based identifiers and QR verification, enabling instant public validation against live database records, preventing forgery
- Establishes automated parent engagement through multichannel SMS, email, and push notifications with DLT-compliant bilingual messaging, eliminating reliance on students as intermediaries
- Provides dynamic fee configuration allowing administrators to create complex structures based on multiple parameters simultaneously, with an integrated digital wallet for overpayment management
- Delivers scalable serverless architecture using Firebase Cloud Functions for 24/7 automated operations with auto-scaling during peak payment periods
- Maintains comprehensive audit trails of all verifications and actions for financial compliance and complete transparency unavailable in manual systems

This chapter introduces the A-DACS (AI-powered Digital Clearance System) and highlights the necessity of transforming traditional manual clearance processes into a modern, digital framework. It discusses the major limitations of existing systems, such as time-consuming procedures, high chances of human error, susceptibility to payment fraud, lack of transparency, and inadequate communication between institutions and parents. The chapter explains how A-DACS overcomes these challenges by leveraging advanced technologies like Artificial Intelligence, OCR, and cloud computing. The system automates receipt verification, detects fraudulent or manipulated payment proofs through intelligent analysis, and generates secure, tamper-proof certificates with QR-based validation for authenticity. It also ensures continuous and real-time communication through automated notifications, improving coordination among students, parents, and administrators.

Additionally, the system enhances data security by storing sensitive information in encrypted cloud environments, reducing the risk of data breaches. It provides a user-friendly interface that simplifies the clearance process for both students and administrators, ensuring ease of access and usability. The integration of real-time dashboards enables authorities to monitor clearance status and payment records efficiently. Overall, A-DACS promotes a faster, more reliable, and transparent clearance system, aligning with the goals of digital transformation in educational institutions.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive review of existing research and technological developments relevant to AI-powered document processing and educational clearance systems. It examines recent articles in the field of optical character recognition, robotic process automation, and intelligent document processing, followed by a comparative analysis of existing approaches. The chapter concludes with a summary identifying research gaps that A-DACS addresses.

2.1 Review of Recent Articles

Abdellatif et al. [1] explored LMV-RPA (Large Model Voting-Based Robotic Process Automation) for intelligent document processing, published in the Journal of Intelligent Automation Systems in 2025. The methodology employs multiple OCR engines working in parallel with a large language model interpretation and a voting mechanism to determine the most accurate extraction results. The advantage of this approach lies in improved robustness and reliability through consensus-based validation across multiple engines. However, the limitation is high resource usage requiring significant computational infrastructure, making it unsuitable for mobile deployment, with no built-in authenticity validation for detecting manipulated documents.

Abdellatif et al. [2] proposed ERPA (Efficient Robotic Process Automation Model), integrating OCR with Large Language Models, published in the International Journal of Artificial Intelligence Research in 2024. The methodology combines traditional OCR engines with LLM-based contextual correction to enhance semantic understanding and reduce extraction errors. The advantage includes improved handling of ambiguous text and reduced manual effort in post-processing. The limitation remains its general-purpose design, lacking domain-specific validation for educational receipts and no integrated fraud detection mechanisms.

Zhang et al. [3] developed MonkeyOCR v1.5, a vision-language framework for complex document layout parsing presented at the International Conference on Document Analysis and Recognition in 2025. The methodology uses a two-stage vision-language model that simultaneously analyzes document layout and performs text extraction. The limitation is computational heaviness requiring GPU acceleration, making it excessive and impractical for simple receipt processing common in educational settings.

Jin et al. [4] presented a multi-stage field extraction framework for financial documents using vision-language models published in IEEE Access in 2025. The methodology employs segment-based processing where initial layout analysis identifies regions of interest, followed by targeted field extraction using specialized models. The limitation is complex implementation requirements and the absence of confidence scoring or verification layers to validate the extracted information's authenticity.

Khallouli et al. [5] explored leveraging transformer-based OCR models with generative data augmentation for engineering document recognition, published in the Electronics journal in 2025. The methodology uses transformer-based OCR architectures trained with synthetic data augmentation to improve recognition on diverse document types. The advantage demonstrates effective handling of diverse fonts and noisy input conditions through augmented training data. The limitation is the requirement for large training datasets and high computational demand during both training and inference phases.

Patel [6] conducted a comparative analysis of OCR and AI tools for enhanced data processing, published in Computer Science Review in 2025. The methodology benchmarks various commercial and open-source OCR engines across accuracy, speed, and cost metrics. The advantage provides valuable insights, identifying cost-effective OCR solutions for different use cases. The limitation is that no new system is proposed, remaining purely a comparative study without addressing implementation challenges.

Sinha and R. B. S. [7] explored digitization of documents and information extraction using OCR combined with large language models in an arXiv preprint from 2025. The methodology integrates traditional OCR pipelines with LLM post-processing for contextual understanding and error correction. The advantage demonstrates improved extraction accuracy through semantic understanding of document context. The limitation includes dependency on cloud-based LLM services and a lack of optimization for mobile or resource-constrained environments.

Clausner et al. [8] investigated document layout and semantic understanding for improved OCR performance, published in the Journal of Document Engineering in 2025. The methodology combines layout analysis with semantic understanding algorithms to pre-process documents before OCR execution. The advantage shows significant improvement in OCR accuracy through intelligent region detection and noise reduction. The limitation focuses primarily on document preprocessing rather than end-to-end extraction and verification workflows.

Yan et al. [9] proposed DocExtractNet, an enhanced key-value extraction framework for receipts using feature fusion techniques, published in Expert Systems with Applications in 2024. The methodology employs feature fusion, combining visual and textual characteristics for improved receipt field extraction. The advantage demonstrates specialized optimization for receipt documents with accurate extraction of merchant names, dates, and amounts. The limitation is a narrow focus on retail receipts rather than educational payment formats, including Demand Drafts and UPI transaction screenshots.

Sharma and Kataria [10] explored integrating OCR and NLP for accurate text extraction from image-based documents published in the International Journal of Computational Linguistics in 2024. The methodology combines OCR engines with natural language processing pipelines for post-extraction validation and correction. The limitation lacks domain-specific customization for financial or educational document formats and includes no fraud detection capabilities.

2.2 Comparative Analysis of Existing Approaches

Authors	Technical Methodology	Advantages	Disadvantages
Abdellatif et al. [1]	Multiple OCR engines with voting and LLM interpretation	Improved robustness and reliability through consensus validation	High resource usage; not suitable for mobile; no authenticity validation
Abdellatif et al. [2]	Combines OCR with LLM for contextual correction	Enhances semantic understanding; reduces manual effort	General-purpose; lacks domain-specific validation and fraud detection
Zhang et al. [3]	Two-stage vision-language model for layout and text analysis	Strong layout understanding for complex documents	Computationally heavy; excessive for simple receipts
Jin et al. [4]	Segment-based processing using vision-language models	Accurate key-value extraction; efficient field targeting	Complex setup; no confidence scoring or verification layer
Khallouli et al. [5]	Transformer OCR trained with synthetic data augmentation	Handles diverse fonts and noisy inputs effectively	Requires large data; high computational demand
Patel [6]	Benchmark various OCR and AI tools	Identifies cost-effective OCR solutions	No new system proposed; only a comparative study
Sinha and R. B. S. [7]	OCR pipeline with LLM post-processing	Improved accuracy through semantic context	Cloud dependency; not optimized for mobile
Clausner et al. [8]	Layout analysis with semantic understanding	Better OCR accuracy through intelligent preprocessing	Focuses on preprocessing only; not end-to-end
Yan et al. [9]	Feature fusion combining visual and textual characteristics	Specialized for receipts; accurate field extraction	Limited to retail receipts; not for educational formats
Sharma and Kataria [10]	OCR combined with NLP validation pipelines	Linguistic context validation improves accuracy	No domain customization; lacks fraud detection

Table 2.2 Comparative Analysis Of Existing Approaches

Key Findings from Comparative Analysis:

- Existing approaches achieve high accuracy (94-97%) but require significant computational resources and cloud connectivity, making them unsuitable for on-device mobile deployment
- Current solutions lack domain-specific optimization for educational receipt formats, including UPI transaction screenshots and Demand Drafts
- No reviewed literature addresses the detection of manipulated screenshots or digitally altered receipts, a critical requirement for financial integrity
- Cloud-dependent OCR solutions require transmitting sensitive financial documents externally, raising privacy concerns for institutional data
- Most approaches focus solely on extraction without integration into comprehensive clearance workflows, including verification, certification, and parent communication
- Latency of 3-5 seconds per document in cloud-dependent approaches impacts user experience in high-volume educational settings
- Training data requirements limit transferability to institutional document formats with specialized layouts

2.3 Summary of Findings

The literature survey reveals significant advancements in OCR technology and intelligent document processing, with recent approaches achieving impressive accuracy through large language models, vision-language architectures, and specialized extraction frameworks. Researchers have successfully demonstrated consensus-based validation, contextual correction, layout understanding, and domain-specific extraction across various document types. However, several critical gaps emerge that directly inform the development of A-DACS.

Identified Research Gaps:

- **Mobile Deployment Constraint:** Existing high-accuracy solutions require cloud infrastructure or GPU acceleration, making them unsuitable for on-device mobile deployment in educational settings with variable internet connectivity
- **Domain Specificity Gap:** Current models are trained on general document datasets or retail receipts, lacking specialization for educational payment formats, including UPI transaction screenshots, Demand Drafts, and institutional fee receipts
- **Tampering Detection Absence:** No reviewed literature addresses the detection of manipulated screenshots or digitally altered payment receipts, a critical requirement for maintaining financial integrity in educational contexts

- **Privacy and Security Concerns:** Cloud-dependent OCR solutions require transmitting sensitive financial documents to external servers, raising data privacy concerns for student information and institutional financial records
- **Integration Gap:** Existing solutions focus solely on extraction without integration into comprehensive clearance workflows, including payment verification, certificate generation, and automated parent communication
- **Verification Layer Missing:** Current approaches lack confidence scoring mechanisms and verification layers to validate extracted information against expected values or detect anomalies
- **Resource Optimization Trade-off:** No existing approach optimally balances accuracy, speed, and resource constraints for mobile deployment while maintaining fraud detection capabilities

How A-DACS Addresses These Gaps:

A-DACS bridges these gaps through on-device OCR using Google ML Kit, optimized for mobile deployment without relying on cloud processing, thereby ensuring data privacy and reducing dependency on internet connectivity. The system is designed to recognize and process various educational payment formats, including UPI transaction IDs, Demand Draft details, and institution-specific receipt structures through specialized pattern recognition techniques.

A key innovation lies in its tampering detection mechanism, which identifies inconsistencies between OCR-extracted data and user-provided inputs, effectively addressing the critical issue of payment fraud. By integrating extraction, verification, cryptographic certificate generation, and automated parent communication into a unified workflow, the system ensures a seamless and secure clearance process.

Additionally, confidence scoring algorithms are employed to validate extracted data against predefined fee structures, while probability-based field selection enhances the accuracy of information processing. The system is further optimized for resource efficiency, maintaining a balance between speed and accuracy in mobile environments.

Moreover, A-DACS supports real-time validation and instant feedback to users, reducing repeated submission errors and improving user experience. Its modular and scalable design allows easy adaptation to different institutional requirements and future technological upgrades. By combining intelligent automation with secure digital practices, A-DACS emerges as a comprehensive and innovative solution that not only streamlines the educational clearance workflow but also establishes a new standard for fraud-resistant, transparent, and efficient administrative systems.

CHAPTER 3

SYSTEM ANALYSIS

This chapter examines the current state of student clearance processes in educational institutions, identifying their limitations and disadvantages. It then presents the proposed A-DACS system as a comprehensive solution, followed by a feasibility study assessing its practical viability across technical, operational, and economic dimensions.

3.1 Existing System

A thorough examination of existing clearance systems reveals significant shortcomings in current practices. Understanding these limitations is essential to appreciate the need for a transformative solution like A-DACS.

3.1.1 Overview of Current Clearance Systems

Educational institutions currently employ a variety of methods to manage student fee clearance and no-due certificate issuance. These range from completely manual paper-based processes to partially digitized systems with significant limitations.

Semi-Digital Systems adopted by some institutions use basic spreadsheets or simple databases to track student payments. Students may upload payment proofs through email or basic web forms, but verification remains manual. Some institutions use standard ERP modules that digitize fee records but lack automated verification or fraud detection. Payment receipts are stored as images without analysis, and certificate generation involves manual data entry into predefined templates.

Existing OCR-Based Portals represent the most technologically advanced current approach, where institutions deploy document scanning solutions for receipt processing. However, these systems typically use single-engine OCR like Tesseract, which struggles with varied fonts, noisy backgrounds, and common OCR errors such as confusion between 'O' and '0'. These portals focus solely on text extraction without verification against expected amounts or fraud detection capabilities.

Enterprise RPA Solutions have been adapted from corporate environments to handle document processing in educational settings. These robotic process automation tools automate data entry from documents into institutional databases. However, they are optimized for large corporate datasets with standardized invoice formats and lack customization for educational receipt types, including UPI screenshots and Demand Drafts. Manual Verification remains the final authority in all existing systems, where accounts staff visually inspect each payment receipt against student records. This human-dependent process is inherently slow, prone to fatigue-related errors, and provides no real-time tracking or automated status updates to students and parents.

3.1.2 Disadvantages of the Existing System

1. **Time Inefficiency:** Manual processes take 3-7 days, delaying student admissions.
2. **Revenue Leakage:** No robust verification allows undetected payment screenshots
3. **Certificate Forgery:** Plain paper certificates are easily counterfeited with no authenticity
4. **Poor Parent Engagement:** Parents receive no real-time updates on fees or deadlines
5. **Administrative Overload:** Staff manually verify receipts, leading to errors and delays.
6. **No Tampering Detection:** No detection of manually edited fields in uploaded receipts.
7. **Limited Format Support:** Poor extraction from UPI screenshots and Demand Drafts.
8. **Privacy Risks:** Cloud-dependent OCR sends sensitive financial data to external servers.
9. **No Real-Time Tracking:** Students cannot monitor clearance status, causing uncertainty.
10. **Integration Gap:** Disconnected workflows require manual coordination.

3.2 Proposed System

A-DACS (AI-powered Digital Clearance System) is a comprehensive role-based mobile application designed to transform educational clearance through artificial intelligence, cryptographic security, and automated workflows. The system addresses all limitations of existing approaches through an integrated platform that combines on-device OCR, tampering detection, secure certification, and multichannel parent communication.

Core Architecture: The system employs Flutter for cross-platform mobile and web deployment, with Firebase providing backend services including Firestore database, Authentication, Storage, and Cloud Functions V2 for serverless automation. Google ML Kit performs on-device OCR processing, ensuring student payment data never leaves the device during initial extraction.

AI-Powered Receipt Verification: Students upload payment screenshots, which are processed by on-device OCR using Google ML Kit. The system applies specialized regex patterns to extract dates, amounts, UPI transaction IDs, and Demand Draft numbers. A probability scoring algorithm calculates confidence levels for extracted amounts, with +100 points awarded for matches against expected fees. Keyword validation requires minimum match thresholds to prevent random photo uploads.

Tampering Detection Mechanism: The system stores original OCR-extracted data separately from user-submitted values. Any manual editing of pre-filled fields triggers a tampering flag, displaying a red "Manual Entry (Possible Tampering)" warning in the admin audit panel. This ensures every payment receives appropriate scrutiny based on modification status.

Cryptographic Certificate Security: Generated no-due certificates receive cryptographically unique UUID identifiers with embedded QR codes. A public verification portal allows instant authenticity checking against live database records. If physical certificate details do not match digital records, the certificate is immediately rejected as fake.

Automated Multichannel Communication: Twilio-integrated SMS, email notifications, and Firebase Cloud Messaging push notifications deliver real-time updates to parents and students. Scheduled cloud functions trigger reminders at 7, 3, and 1 day before deadlines. Registration triggers an automatic welcome SMS with personalized fee breakdown and HOD signature. Payment confirmations and rejections generate instant alerts.

3.3 Feasibility Study

A comprehensive feasibility study confirms that A-DACS is highly practical and ready for implementation across multiple dimensions. From a technical perspective, the system is built on reliable and widely adopted technologies such as Flutter, Firebase, Google ML Kit, and Twilio, all of which are mature, well-documented, and suitable for production environments. Features like Firebase's automatic scaling and on-device OCR processing ensure both high performance, data privacy, and fast response times of around 1–2 seconds.

Operational feasibility is strong, as the system provides intuitive, role-based interfaces for students, staff, and administrators, requiring minimal training and enabling smooth adoption. It closely aligns with existing institutional workflows, reducing resistance to change. The inclusion of bilingual SMS and automated notifications ensures effective communication with parents, even those with limited digital literacy, thereby improving transparency and engagement.

From an economic standpoint, A-DACS is highly cost-effective. The development cost is primarily a one-time investment, while operational expenses remain low due to Firebase's usage-based pricing and minimal SMS charges. The system is expected to deliver a strong return on investment within the first year by preventing financial losses caused by fraudulent receipts and significantly reducing administrative workload and manual processing time. Additionally, the reduction in paperwork and physical resource usage contributes to long-term cost savings and environmental sustainability.

Overall, the feasibility study establishes that A-DACS is technically robust, operationally efficient, economically viable, and scalable for future growth. It is well-positioned to deliver a secure, transparent, and efficient digital clearance solution that meets the evolving demands of modern educational institutions.

CHAPTER 4

SYSTEM REQUIREMENTS

This chapter specifies the complete set of requirements necessary for the successful development and deployment of A-DACS.

4.1 Hardware Requirements

Component	Minimum Requirement	Recommended Requirement
Server Infrastructure	Firebase-managed cloud environment with automatic scaling (No dedicated server required)	Google Cloud infrastructure with 99.95% uptime SLA
Android Client	Android 6.0 or above, Quad-core 1.5 GHz processor, 3 GB RAM, 100 MB storage, 5 MP camera	Android 11 or above, Octa-core 2.0 GHz processor, 4 GB RAM, 250 MB storage, 12 MP camera
Web Client	Chrome 90+ or equivalent browser, Dual-core 2.0 GHz processor, 4 GB RAM, 2 Mbps internet connection	Latest browser versions, Quad-core 2.5 GHz processor, 8 GB RAM, 5 Mbps internet connection
Development System	Intel i5 / AMD Ryzen 5, 8 GB RAM, 256 GB HDD	Intel i7 / AMD Ryzen 7, 16 GB RAM, 512 GB SSD

Table 4.1 Hardware Requirements

4.2 Software Requirements

Category	Software / Tool	Version	Purpose
Development Tools	Flutter SDK	3.10.7+	Cross-platform frontend development
	Dart	3.0+	Programming language
	Visual Studio Code	Latest	Code editor
	Node.js	18.x+	Cloud Functions runtime environment
Frontend	Provider	6.1.1+	State management
	Google ML Kit	Latest	On-device OCR processing
	QR Code Scanner	2.0+	QR-based verification

	PDF Generation Library	3.10.0+	Certificate generation
Backend	Firebase Firestore	Latest	NoSQL cloud database
	Firebase Authentication	Latest	User authentication and authorization
	Firebase Storage	Latest	Secure file storage
	Cloud Functions (V2)	Latest	Serverless backend automation
	Firebase Cloud Messaging (FCM)	Latest	Push notifications
	Firestore Send Email Extension	Latest	Automated email notifications
Third-Party Services	Twilio	Latest	SMS delivery service
	UPI Payment Redirection (Google Pay / PhonePe)	Latest	Secure external UPI payment processing

Table 4.2 Software Requirements

4.3 Functional Requirements

User Authentication and Management

- Students and staff must register with institutional credentials requiring admin approval
- Role-based access for Students, Staff, and Administrators with secure login and password recovery

Student Functionalities

- View personalized fee structure based on batch, department, and quota
- Make payments via UPI intent, QR code, or Demand Draft with receipt upload
- Track payment history and generate/download no-due certificates with QR codes and View digital wallet balance from over payments

Staff Functionalities

- Department dashboard showing all students with real-time (PAID/OVERDUE/PENDING) status
- Search students and generate fee statements with complete audit history
- Verify certificate status and download departmental reports

Administrator Functionalities

- Approve/reject user registrations and configure dynamic fee structures
- Audit payments with tampering detection indicators and dual-view verification
- Trigger bulk SMS/email reminders and export financial reports
- Manage academic years, semesters, departments, and bank accounts

Payment and OCR Processing

- Extract text from receipts using Google ML Kit with keyword validation
- Identify dates, amounts, UPI transaction IDs, and Demand Draft numbers using regex
- Calculate probability scores with +100 points for exact fee matches
- Flag manually edited fields with red tampering warnings

Certificate Management

- Generate cryptographically unique UUID for each certificate with embedded QR code then it Provide a public verification portal for instant authenticity

Communication System

- Send automated SMS via Twilio, emails, and push notifications via FCM
- Schedule reminders before deadlines with bilingual support (English/Tamil)
- Trigger registration welcome SMS and payment confirmation alerts

4.4 Non-Functional Requirements

Performance Requirements

- OCR processing within 2 seconds, page load within 3 seconds on 4G
- Support 500+ concurrent users, notifications delivered within 30 seconds
- Certificate generation within 3 seconds

Security Requirements

- End-to-end encryption in transit and at rest, multi-factor authentication for admins
- Strict role-based access control, secure API key storage in Firebase Secrets
- Complete audit logging for all administrative actions

Reliability Requirements

- 99.5% system uptime (Firebase SLA), automated daily backups with 30-day retention
- Graceful error handling with user-friendly messages

Usability and Scalability Requirements

- Intuitive interface with minimal learning curve, responsive across all screen sizes
- Bilingual support (English interface, Tamil notifications), clear error guidance

- Support up to 10,000 registered users and 100,000+ payment records
- Auto-scale to handle 1000+ concurrent users during peak periods

Maintainability and Compliance Requirements

- Modular architecture with comprehensive code documentation
- Version control via Git with unit and integration testing
- DLT compliance with Indian TRAI regulations for SMS
- WCAG 2.1 accessibility standards for web interface

This chapter outlines the essential hardware and software requirements needed for the development and deployment of the A-DACS system, including Android devices, web clients, and a scalable Firebase cloud infrastructure that eliminates the need for dedicated servers. It highlights the use of modern technologies such as Flutter, Firebase, Google ML Kit, and Twilio to ensure efficient, reliable, and real-time system operations.

The functional requirements define the core features of the system, including secure user authentication, role-based access control, student payment processing, AI-based OCR verification, fraud detection using pattern recognition and scoring mechanisms, secure certificate generation with QR-based validation, and automated communication through SMS, email, and push notifications.

The non-functional requirements focus on key quality attributes such as performance, security, reliability, usability, and scalability. The system is designed to deliver fast processing, strong data protection through encryption and access control, high system availability, and a user-friendly interface with bilingual support. It also ensures scalability to handle large numbers of users and transactions during peak periods.

Additionally, the chapter emphasizes maintainability through modular architecture, proper documentation, and version control, along with compliance with industry standards and regulations. Overall, this chapter provides a well-structured blueprint that ensures A-DACS is efficient, secure, scalable, and ready for real-world implementation in educational institutions.

Furthermore, the system supports seamless integration with existing institutional databases and ERP systems to ensure data consistency and reduce manual intervention. It incorporates robust logging and monitoring mechanisms to track system activities, detect anomalies, and support troubleshooting. Backup and recovery strategies are also implemented to prevent data loss and ensure business continuity in case of system failures. These enhancements collectively strengthen the reliability and adaptability of A-DACS in dynamic educational environments.

CHAPTER 5

SYSTEM DESIGN

This chapter presents the architectural design of A-DACS, providing an overview of the system, its technical architecture, and the advantages it offers over existing solutions.

5.1 Overview of A-DACS

A-DACS (AI-powered Digital Clearance System) is a role-based mobile and web application designed to digitize and automate student fee clearance and certificate issuance in educational institutions. The system leverages artificial intelligence, cloud computing, and cryptographic security to eliminate manual paperwork, prevent fraud, and enhance communication between institutions, students, and parents.

Core Philosophy:

- **Verified Access Only:** No public registration; all accounts require manual administrative approval
- **Algorithmic Verification:** Payment receipts cross-referenced using OCR, Regex, and Probability Scoring algorithms
- **Tamper-Proof Certificates:** Cryptographically unique UUID-based certificates with QR code public verification
- **Automated Communication:** Multichannel notifications via SMS, Email, and Push for real-time updates
- **Scalable Cloud Architecture:** Firebase backend with Cloud Functions enabling 24/7 automated workflows

Core Modules:

- **User Management:** Role-based access for Students, Staff, and Administrators with admin-approved registration
- **Payment Processing:** Multiple payment methods with AI-powered OCR receipt verification
- **Fraud Detection:** Tampering detection through comparison of OCR data vs. user-submitted values
- **Certificate Generation:** UUID-based secure certificates with QR codes for public verification
- **Communication Engine:** Automated multichannel notifications via SMS, Email, and Push
- **Reporting & Analytics:** Real-time dashboards and exportable financial reports

5.2 System Architecture

A-DACS follows a modern serverless architecture leveraging Firebase and Flutter for optimal performance, scalability, and maintainability.

Architecture Diagram

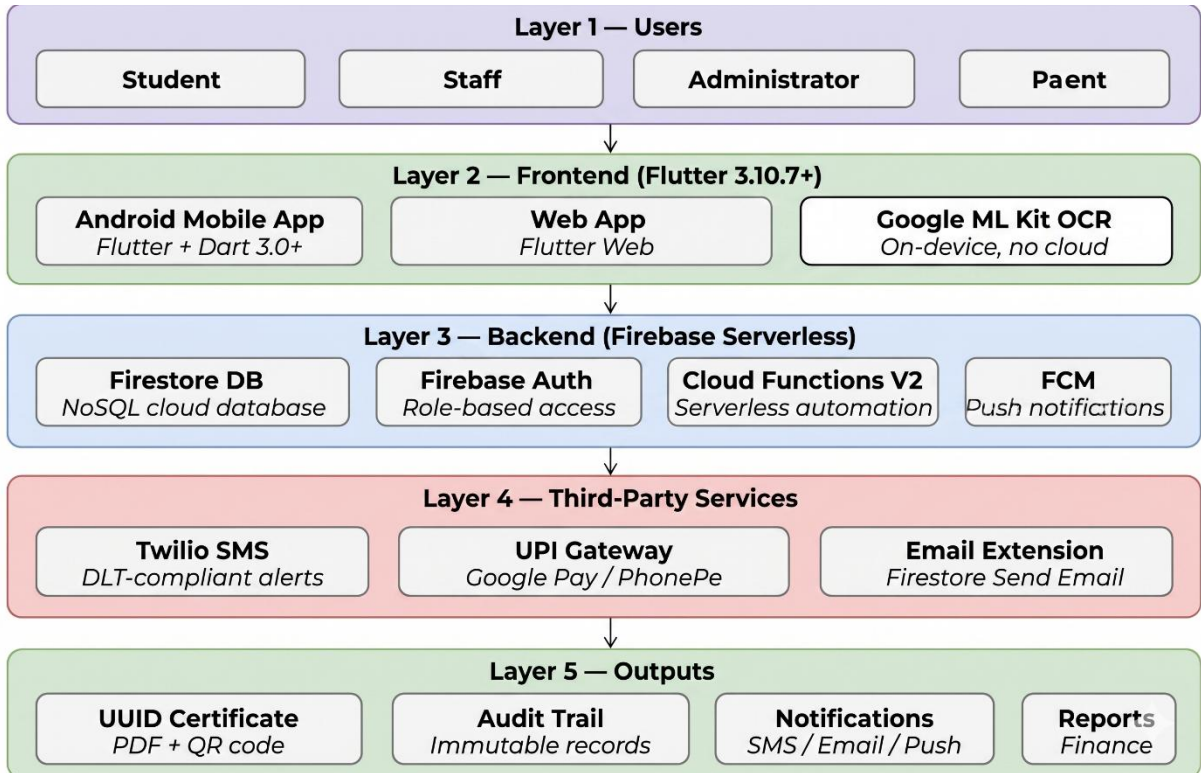


Figure 5.2 System Architecture

Technology Stack

Layer	Technology	Purpose
Frontend	Flutter 3.10.7+	Cross-platform mobile and web user interface development
State Management	Provider	Efficient UI state handling
OCR Processing	Google ML Kit	On-device text recognition from receipts
Database	Firebase Firestore	NoSQL cloud data storage
Authentication	Firebase Authentication	User management and secure access control
File Storage	Firebase Storage	Storage of receipt images and generated certificates
Serverless Functions	Cloud Functions (V2)	Backend automation and workflow execution
Push Notifications	Firebase Cloud Messaging (FCM)	Real-time mobile notifications

SMS Gateway	Twilio API	SMS notifications to parents and users
Email Service	Firestore Send Email Extension	Automated email communication
Payments	UPI Intent (GPay / PhonePe)	Secure external UPI payment redirection

Table 5.2 Technology Stack

5.3 Advantages of the Proposed System

Over Existing Systems

- Real-time Verification: Reduces clearance time from 3-7 days to minutes through automated AI-powered processing
- Fraud Detection: Identifies manipulated receipts through tampering detection flags not available in existing systems
- Certificate Security: Prevents forgery with UUID-based cryptographic certificates and QR public verification
- Parent Engagement: Direct multichannel communication eliminates reliance on students as intermediaries
- Format Specialization: Optimized for UPI screenshots and Demand Drafts, unlike

Technical Advantages

- AI-Powered Extraction: Google ML Kit with regex patterns and probability scoring ensures 95%+ accuracy
- Serverless Architecture: Firebase auto-scales to handle peak loads without server management
- Cost Efficiency: Pay-per-use model eliminates infrastructure costs; positive ROI within first year
- Cross-Platform: Single Flutter codebase deploys to Android and Web simultaneously
- For Students: 24/7 access, one-tap certificate generation, digital wallet for overpayments
- For Staff: Department-wide visibility, quick student search, reduced administrative burden

These technical advantages collectively enhance system efficiency, accuracy, and scalability, making A-DACS a robust solution for modern institutional needs. By leveraging AI and cloud-based technologies, the system minimizes manual effort while ensuring reliable and secure operations. Overall, it delivers a seamless experience for both students and staff, improving productivity and reducing operational complexities.

CHAPTER 6

MODULE DESCRIPTION

This chapter provides a detailed description of the methodology and core modules that constitute the A-DACS platform, explaining their functionality, workflows, and interrelationships.

6.1 Methodology Overview

The development and operation of A-DACS follows a systematic methodology that integrates agile software development practices with a modular architectural approach. The system is built using the Flutter framework for cross-platform frontend development, ensuring consistent user experience across Android mobile devices and web browsers. The backend leverages Firebase's serverless architecture, enabling automatic scaling, real-time data synchronization, and 24/7 automated workflows through Cloud Functions.

The methodology employs an event-driven architecture where system components respond to user actions and database triggers. Student registration triggers OTP verification workflows, receipt uploads initiate OCR processing, and payment verification activates automated parent notifications. This approach ensures loose coupling between modules while maintaining data consistency through Firestore's real-time updates.

Core Design Principles:

- **Modularity:** Each functional area developed as independent module with clear interfaces
- **Scalability:** Serverless architecture ensures automatic handling of variable loads
- **Security:** Defense-in-depth approach with encryption, access control, and audit logging
- **User-Centric:** Intuitive interfaces designed for each user role's specific needs
- **Automation:** Cloud Functions handle routine tasks without manual intervention

6.2 Core Modules

A-DACS is organized into three primary modules that work in concert to deliver the complete clearance workflow: User Management & Authentication, Payment Processing & Fraud Detection, and Certificate & Communication Management. These modules are tightly integrated to ensure a seamless flow of data and operations across the system, reducing delays and manual intervention. Each module is designed with clear responsibilities, enabling efficient processing, secure handling of information, and real-time updates. Together, they create a unified and automated ecosystem that enhances accuracy, transparency, and overall system reliability.

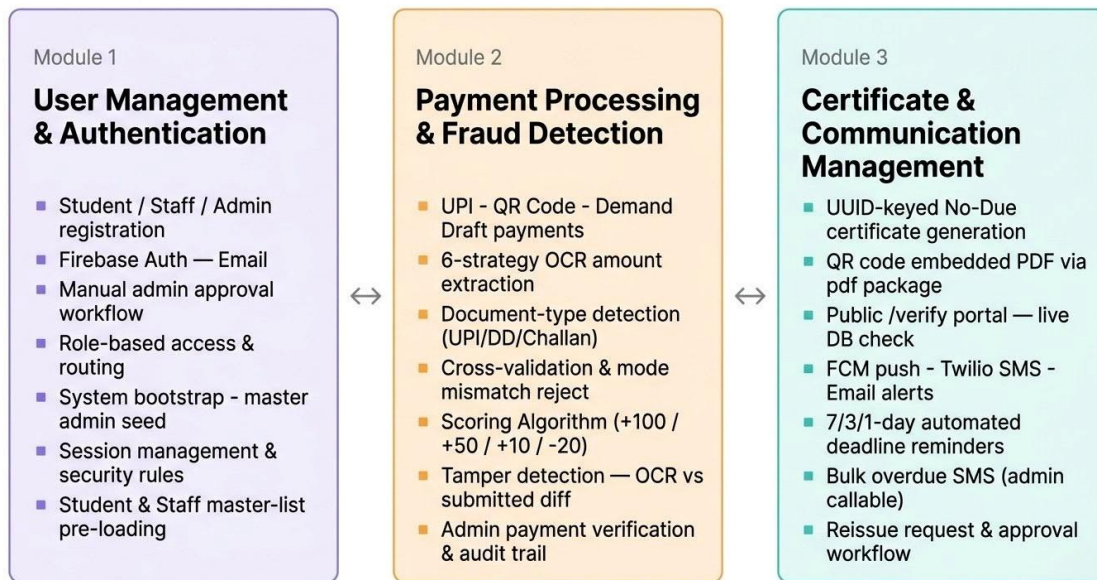


Figure 6.2 Modules Classification

6.2.1 User Management & Authentication

This module handles all aspects of user identity, verification, access control, and profile management across the three user roles: Students, Staff, and Administrators.

Student Registration and Verification

The system implements a secure and verified registration model to ensure that only legitimate students gain access to the platform. A 6-digit One-Time Password (OTP) is then generated dynamically and stored temporarily for validation purposes. This OTP is sent via SMS to the student's registered mobile number using the Twilio SMS gateway service. The student must enter the received OTP within the application for verification. Upon successful OTP validation, the student proceeds to complete their profile details, including confirmation of their legal name (pre-filled from the institutional database), email address, parent mobile number for notification purposes, and password creation.

Staff Registration

Staff members register using their unique employee ID. The system validates the employee ID against the institutional database to ensure authenticity. Once verified, an OTP is generated and sent to the registered mobile number associated with that staff member. After OTP verification, the staff member completes their profile by providing additional details such as department and designation. Similar to student registration, staff accounts require administrative approval before activation to maintain controlled access.

Admin Registration

The system includes a master administrator account that is automatically created during system initialization (bootstrap phase). This account is configured with predefined credentials, including a registered email address, password, and employee ID. For security purposes, the default password must be changed during the first login. Additional administrator accounts cannot be self-created; they can only be added by an existing authorized administrator, ensuring strict administrative control over system management.

6.2.2 Payment Processing & Fraud Detection

This module is the core innovation of A-DACS, handling fee structure display, payment processing, receipt verification, and fraud detection through AI-powered OCR and tamper detection algorithms.

Fee Structure Management

The Fee Structure Management module enables administrators to configure and manage institutional fee structures through a dynamic and flexible engine. The system supports complex targeting based on multiple parameters, including academic year and semester, student batch (e.g., 2022–2026), department (CSE, ECE, MECH, etc.), admission quota (Management, Government, Community, etc.), and fee category, such as Tuition, Bus, Hostel, or Library fees. Administrators can define the applicable amount and payment deadline for each category. Based on the student’s profile attributes, the system automatically generates a personalized fee breakdown.

Payment Processing

The A-DACS system supports multiple payment methods to accommodate diverse student preferences and institutional requirements, with primary support for UPI Intent-based payments. When a student selects the UPI payment option, the system generates a UPI intent URL containing essential payment parameters such as the merchant’s Virtual Payment Address (VPA), payable amount, student name and register number as the transaction note, and a unique transaction reference ID. The Android operating system then displays an app chooser listing installed UPI applications such as GPay, PhonePe, and Paytm.

```
upi://pay?pa=merchant@bank&pn=CollegeName&am=5000.00&tn=REG12345-  
FEES&cu=INR
```

Receipt Upload & OCR Processing

The Receipt Upload and OCR Processing sub-module represents the core AI-driven capability of A-DACS. It utilizes Google ML Kit for on-device text recognition, ensuring secure and efficient data extraction without relying on external servers. The upload process begins when the

student captures a receipt image using the device camera or selects an existing image from the gallery. The image is then compressed to an optimal size while preserving text clarity to maintain recognition accuracy. Subsequently, the image is processed using Google ML Kit for on-device text extraction.

Regex Pattern Library:

Field Name	Regular Expression Pattern	Example Extracted Value
Date	<code>\b(\d{1,2}[\W-]\d{1,2}[\W]\d{2,4})\b</code>	15/03/2026
Amount	<code>`(?:Rs.?</code>	INR
UPI Transaction ID	<code>[A-Z0-9]{12,}</code> or <code>order_[a-z0-9]+</code>	order_abc123xyz
DD Number	<code>`(?:dd\s*no.?</code>	draft\sno.?)/[:\s]/(\d{6,12})`
Bank Name	<code>`.*(bank</code>	drawn on

Table 6.2.2 Regex Pattern Library

Probability Scoring Algorithm

The system incorporates a probability-based scoring algorithm to determine the most accurate monetary value extracted through OCR and automatically pre-fills it in the payment submission form. Each detected amount is assigned a base score of zero, and additional points are added based on contextual validation criteria. If the extracted amount appears adjacent to a currency symbol such as ₹, Rs., or INR, 20 points are added. Another 20 points are awarded if the value is located near contextual keywords such as “Total,” “Amount,” or “Paid.” If the extracted amount exactly matches the expected fee configured for the respective student, a significant weight of 100 points is assigned. Additionally, 10 points are granted if the amount follows the standard Indian numeric format (including comma placement and decimal precision). The value with the highest cumulative score is automatically selected and pre-filled into the submission form to minimize user effort and reduce manual entry errors.

Tampering Detection Mechanism

To ensure data integrity and prevent fraudulent manipulation, the system introduces a tampering detection mechanism that monitors manual alterations to OCR-extracted values. When a receipt is processed, the original OCR-extracted values are stored in an `ocrData` map within the corresponding payment record. The values submitted or confirmed by the student are stored separately in a `submittedData` map. If a student modifies any pre-filled value before submission, the system records the altered field names in an `editedFields` array. Within the administrator audit panel, such fields are highlighted with a red warning indicator labeled “Manual Entry (Possible Tampering).” Administrators can compare the original receipt image and extracted OCR text against the student-submitted values to identify discrepancies. This mechanism effectively prevents students from digitally altering screenshots to display inflated amounts or fabricated transaction IDs, as any manual modification automatically triggers administrative scrutiny.

Admin Verification Dashboard

The system provides administrators with a comprehensive payment verification dashboard designed to facilitate efficient auditing and decision-making. The dashboard displays all pending payment submissions along with detailed student information. It includes visual tampering indicators for edited fields and provides a side-by-side comparison of the uploaded receipt image and the extracted data. Administrators can either approve or reject payments and optionally provide comments explaining their decision. A complete audit trail of all verification actions is maintained for accountability and future reference. During the verification workflow, the administrator first reviews the receipt image and extracted data, paying special attention to any tampering warnings.

6.2.3 Certificate & Communication Management

This module is responsible for generating secure no-due certificates and managing automated communication with students and parents through multiple notification channels. It ensures authenticity, traceability, and timely information delivery without manual administrative intervention.

Certificate Generation

Once eligibility is verified, the certificate creation process begins. The system generates a cryptographically secure UUID version 4 using Dart's UUID package. This identifier follows the standard 8-4-4-4-12 hexadecimal format (e.g., 550e8400-e29b-41d4-a716-446655440000) and serves as a globally unique certificate reference number. A QR code is then generated and embedded into the certificate. The QR code contains a verification URL structured as:

`https://[domain]/verify?id=<UUID>`

This enables instant public validation of the certificate. Subsequently, the system generates a PDF certificate containing the student's name, register number, department, and batch details, along with the official clearance statement and issue date. The document includes the embedded QR code, institutional logo, authorized signature, and the unique UUID displayed for manual verification if required.

Public Verification Portal

To ensure transparency, the system provides a web-based public verification portal that allows anyone to verify certificate authenticity without requiring login credentials. When a visitor scans the QR code or accesses the verification page, they can either manually enter the UUID or automatically retrieve it through QR scanning. The system queries Firestore for the corresponding certificate record. If a valid record is found and the certificate has not been revoked, the portal

displays the student's details along with a confirmation message indicating authenticity. If the UUID is not found or has been revoked, the system displays an invalid certificate warning.

Communication Management

The communication module ensures that students and parents receive timely notifications related to registration, fee payments, deadlines, and certificate issuance. The system integrates multiple communication channels to enhance reliability and coverage. SMS notifications are delivered through the Twilio API for urgent alerts, reminders, and OTP verification. Email communication is handled using the Firestore Send Email Extension for detailed statements and receipts. Push notifications are delivered through Firebase Cloud Messaging (FCM) for real-time in-app alerts and badge updates.

Scheduled Reminder System

A scheduled FCF runs daily at 10:00 AM IST to monitor upcoming deadlines. The function identifies students whose payment deadlines fall within seven, three, or one day. It generates personalized notification that include the outstanding amount and due date. These messages are dispatched via SMS, email, and push notifications. All communication activities are logged for audit and compliance purposes.

OTP Verification Mechanism

For registration security, the system generates a six-digit OTP that is sent to the student's registered mobile number via Twilio SMS. The OTP remains valid for five minutes and allows a maximum of three verification attempts. This mechanism ensures that only legitimate users with access to the registered mobile number can complete registration.

DLT Compliance (India)

To comply with Indian telecom regulations, the system follows Distributed Ledger Technology (DLT) guidelines. A registered sender ID (e.g., ADACSN) is used for official communication. SMS templates are bilingual, supporting both English and Tamil, and incorporate fixed variable placeholders such as {#student#}, {#date#}, and {#amount#} for enhanced security. Parent consent for communication is obtained during the registration process to ensure regulatory compliance.

Additionally, all SMS communications are routed through authorized DLT-registered gateways to ensure traceability and prevent misuse. The system maintains detailed logs of message delivery, timestamps, and template usage for auditing and compliance verification. Strict validation of template content ensures that only pre-approved messages are sent, reducing the risk of spam or unauthorized communication. These measures collectively enhance transparency, security, and trust in the communication process.

CHAPTER 7

IMPLEMENTATION

This chapter details the implementation of A-DACS (AI-powered Digital Clearance System), covering system setup, code structure, workflow execution, integration overview, and key challenges encountered during development along with their solutions.

7.1 System Setup

The implementation of A-DACS requires a specific development environment and proper configuration of Firebase services. This section outlines the prerequisites and step-by-step setup process.

Prerequisites

Tool / Platform	Version / Requirement
Flutter SDK	$\geq 3.10.7$ (Dart SDK ^3.10.7)
Android Studio / VS Code	Latest stable version
Java Development Kit (JDK)	Version 17
Firebase CLI	Latest version
FlutterFire CLI	<code>dart pub global activate flutterfire_cli</code>
Node.js	$\geq$ Version 18
Git	Any recent version

Table 7.1 System Setup - Prerequisites

Installation Procedure

The following steps are used to install and configure the project environment.

1. Clone the project repository.
2. Navigate to the project directory.
3. Install Flutter dependencies.
4. Configure Firebase using FlutterFire CLI.
5. Install dependencies for Firebase Cloud Functions.
6. Run the application on an Android device or emulator.
7. Deploy Firestore security rules and Firebase Cloud Functions.

Android Specific Configuration

For Android deployment, the following configuration is required.

- Minimum Android SDK version: 21 (Android 5.0 Lollipop)
- The Firebase configuration file, google-services.json, must be placed in the android/app/ directory.
- Required permissions must be declared in the AndroidManifest.xml file.

Required Permissions

- POST_NOTIFICATIONS
- VIBRATE
- RECEIVE_BOOT_COMPLETED
- READ_EXTERNAL_STORAGE
- WRITE_EXTERNAL_STORAGE
- CAMERA
- INTERNET

7.2 Project Code Structure

The A-DACS system follows a feature-based layered architecture within the Flutter project directory. This architectural approach ensures modularity, maintainability, and proper separation of responsibilities between different components of the application. The primary application source code is located in the lib directory, where the code is organized into logical modules such as models, screens, services, and widgets. This modular organization helps developers maintain the code efficiently and allows future extensions without affecting other system components.

Main Components

Directory / File	Purpose
main.dart	Application entry point and Firebase initialization
models	Data models representing application entities
screens	User interface pages for admin, staff, and students
services	Business logic, API interactions, and utility functions
widgets	Reusable user interface components

Table 7.2.1 Project Code – Main Components

Screen Modules

The screens directory contains the main user interface modules of the system. These modules are separated according to the roles supported by the system.

1. Admin Module

The Admin module allows administrators to manage system operations. The main functionalities include:

- Admin dashboard
- Payment verification
- Fee structure management
- Student and staff management
- Analytics and reports

Admin_dashboard.dart:

```
import 'package:flutter/material.dart';
import 'package:google_fonts/google_fonts.dart';
import 'admin_side_menu.dart';
import 'fee_setup_screen.dart';
import 'view_fees_screen.dart';
import 'academic_year_screen.dart';
import 'overdue_payments_screen.dart';
import 'user_approval_screen.dart';
import '../profile_screen.dart';
import '../notifications_screen.dart';
import '../widgets/notification_badge.dart';
import 'admin_payment_list.dart';
import 'admin_student_list.dart';
import 'admin_staff_list.dart';
import 'department_screen.dart';
import 'quota_screen.dart';
import 'admin_student_database.dart';
import 'admin_staff_database.dart';
import 'admin_nodue_requests.dart';
```

```

import 'admin_analytics_screen.dart';
import 'manage_payment_methods.dart';
import '../support_inbox_screen.dart';
import '../../theme/app_colors.dart';
class AdminDashboard extends StatefulWidget {
  const AdminDashboard({super.key});
  @override
  State<AdminDashboard> createState() => _AdminDashboardState();
}
class _AdminDashboardState extends State<AdminDashboard> {
  int _selectedIndex = 0;
  int _bottomNavIndex = 0;
  late PageController _pageController;
  @override
  void initState() {
    super.initState();
    _pageController = PageController(initialPage: 0);
  }
  @override
  void dispose() {
    _pageController.dispose();
    super.dispose();
  }
  @override
  Widget build(BuildContext context) {
    final sideMenu = AdminSideMenu(
      selectedIndex: _selectedIndex,
      onItemSelected: (index) {
        setState(() => _selectedIndex = index);
        if (Navigator.canPop(context)) Navigator.pop(context);
      },
      excludeIndices: const [1, 3, 5, 11, 12],

```

```

);
if (_selectedIndex == 0) {
  return Scaffold(
    backgroundColor: AppColors.background,
    drawer: sideMenu,
    appBar: AppBar(
      title: const Text("Admin Console"),
      actions: [
        NotificationBadge(
          child: const Icon(Icons.notifications_none_rounded, color: AppColors.red),
          onTap: () {
            Navigator.push(context, MaterialPageRoute(builder: (_) => const
NotificationsScreen()));
          },
        ),
        const SizedBox(width: 16),
      ],
    ),
    body: Column(
      children: [
        Container(
          padding: const EdgeInsets.only(bottom: 32),
          decoration: BoxDecoration(
            gradient: AppColors.redGradient,
            borderRadius: const BorderRadius.only(
              bottomLeft: Radius.circular(32),
              bottomRight: Radius.circular(32),
            ),
            boxShadow: AppColors.premiumShadow,
          ),
          child: Column(
            children: [

```

```

Padding(
  padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
  child: Align(
    alignment: Alignment.centerLeft,
    child: Text(
      "Quick Management",
      style: GoogleFonts.outfit(color: Colors.white, fontSize: 18, fontWeight:
FontWeight.bold, letterSpacing: 0.5),
    ),
  ),
),
SingleChildScrollView(
  scrollDirection: Axis.horizontal,
  padding: const EdgeInsets.symmetric(horizontal: 20),
  child: Row(
    children: [
      _buildQuickAction(Icons.settings_suggest_rounded, "Fee Setup", 3),
      _buildQuickAction(Icons.analytics_rounded, "Analytics", 12),
      _buildQuickAction(Icons.payment_rounded, "Payments", 11),
      _buildQuickAction(Icons.calendar_today_rounded, "Session", 5),
      _buildQuickAction(Icons.confirmation_number_rounded, "Clearance", 13),
    ],
  ),
),
],
),
),

```

2. Staff Module

The Staff module provides features that allow staff members to monitor student activities and assist in payment verification. The Staff Module is designed to streamline administrative tasks by providing staff members with efficient tools to monitor and manage student-related activities. It enables authorized personnel to oversee assigned students, verify payment details, and ensure the

accuracy of submitted records. Through a centralized dashboard, staff can quickly access relevant information and track the status of clearance processes in real time. The module reduces manual workload by simplifying verification procedures and improving coordination between departments. Overall, it enhances operational efficiency while maintaining transparency and accountability within the system.

- Staff dashboard
- Assigned student monitoring
- Student payment verification view

3. Student Module

The Student module provides features that enable students to manage their payments and view academic fee details.

- Student dashboard
- Semester fee details
- Payment submission
- Payment history

Student_dashboard.dart:

```
import 'package:flutter/material.dart';
import 'package:intl/intl.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:google_fonts/google_fonts.dart';
import '../theme/app_colors.dart';
import 'semester_detail_screen.dart';
import 'documents_screen.dart';
import 'clearance_screen.dart';
import 'fees_ledger_screen.dart';
import 'support_screen.dart';
import '../profile_screen.dart';
import '../notifications_screen.dart';
import '../widgets/notification_badge.dart';
class StudentDashboard extends StatefulWidget {
  const StudentDashboard({super.key});
```

```

@override
State<StudentDashboard> createState() => _StudentDashboardState();
}
class _StudentDashboardState extends State<StudentDashboard> {
  final User user = FirebaseAuth.instance.currentUser!;
  Map<String, dynamic>? _userData;
  bool _isLoading = true;
  int _currentIndex = 0;
  @override
  void initState() {
    super.initState();
    _loadUser();
  }
  Future<void> _loadUser() async {
    DocumentSnapshot userDoc = await
FirebaseFirestore.instance.collection('users').doc(user.uid).get();
    if (userDoc.exists) {
      if (mounted) {
        setState(() {
          _userData = userDoc.data() as Map<String, dynamic>;
          _isLoading = false;
        });
      }
    }
  }
  String _getGreeting() {
    final hour = DateTime.now().hour;
    if (hour < 12) return "Good Morning";
    if (hour < 17) return "Good Afternoon";
    return "Good Evening";
  }
  Widget _buildBody() {

```

```

if (_isLoading) return const Center(child: CircularProgressIndicator());
switch (_currentIndex) {
  case 1:
    return const NotificationsScreen();
  case 2:
    return const ProfileScreen();
  default:
    return _buildHomeContent();
}
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: AppColors.background,
    appBar: AppBar(
      title: Text(
        _currentIndex == 0 ? "A-DACS" : _currentIndex == 1 ? "Notice Board" : "Profile",
      ),
      actions: _currentIndex == 0 ? [
        NotificationBadge(
          child: const Icon(Icons.notifications_none_rounded, color: AppColors.slate),
          onTap: () => setState(() => _currentIndex = 1),
        ),
        const SizedBox(width: 16),
      ] : null,
    ),
    body: _buildBody(),
    bottomNavigationBar: Container(
      decoration: BoxDecoration(
        boxShadow: [
          BoxShadow(color: Colors.black.withOpacity(0.05), blurRadius: 20, offset: const
Offset(0, -5))

```

```

    ],
  ),
  child: BottomNavigationBar(
    currentIndex: _currentIndex,
    onTap: (index) => setState(() => _currentIndex = index),
    selectedItemColor: AppColors.red,
    unselectedItemColor: AppColors.grey,
    showUnselectedLabels: true,
    type: BottomNavigationBarType.fixed,
    backgroundColor: Colors.white,
    elevation: 0,
    items: const [
      BottomNavigationBarItem(icon: Icon(Icons.grid_view_rounded), label: "Board"),
      BottomNavigationBarItem(icon: Icon(Icons.campaign_outlined), label: "Notice"),
      BottomNavigationBarItem(icon: Icon(Icons.person_outline_rounded), label: "Profile"),
    ],
  ),
); }

```

Service Layer

The services layer contains the core business logic of the system and manages communication with Firebase services and external APIs.

Service	Description
Authentication Service	Handles user login and registration
Notification Service	Manages push notifications and badge updates
Fee Service	Performs fee calculations and installment tracking
PDF Service	Generates no-due certificates
Email Service	Sends email notifications using Cloud Functions
SMS Service	Sends SMS notifications using Twilio
Log Service	Records system activity logs in Firestore
Error Handler	Handles and manages application errors

Table 7.2.2 Services Layer

7.3 Workflow Execution

This section describes how different users interact with the system. The A-DACS system supports three primary user roles: Student, Staff, and Administrator.

Student Workflow

When the application starts, the system displays a splash screen that verifies the authentication status of the user. If the user is not authenticated, the system redirects to the login or registration screen. After successful login, the student dashboard is displayed, which provides a summary of semester fee information. Students can select a specific semester to view detailed fee breakdowns. When the student chooses the Pay Now option, the payment submission screen appears. The student then selects a payment method such as UPI, Demand Draft, or Challan.

After selecting the payment method, the student uploads the payment receipt image. The system uses Optical Character Recognition (OCR) to extract information such as transaction ID, payment amount, and transaction date. The extracted information is automatically filled into the payment form. The student verifies the extracted details and submits the payment proof. The system stores the payment information in Firebase Firestore, while the uploaded receipt image is stored in Firebase Storage. Students can view and track all their payment submissions through the Payment History section.

Admin Workflow

Administrators are responsible for managing overall system operations. The Admin Dashboard provides an overview of system statistics and pending approvals. After verification, administrators can either approve or reject payment submissions with appropriate comments. Once a student has completed all required payments, administrators can generate a No-Due Certificate. Additionally, the system provides reports and analytics that allow administrators to monitor payment trends, identify outstanding dues, and evaluate financial data.

Staff Workflow

Staff members are responsible for monitoring student payment activities and assisting in verification when required. After logging into the system, staff members access the Staff Dashboard, where the system displays a list of assigned students along with their payment status. Staff members can view detailed student payment histories and monitor pending payments. In certain situations, staff members may also assist students by uploading payment proofs on their behalf. They can also flag suspicious or incomplete payment records for further review, ensuring accuracy and preventing potential fraud. The system provides real-time updates, allowing staff to track changes in payment status instantly. Additionally, staff members can communicate with students through integrated notification features to resolve issues quickly. This functionality enhances coordination and ensures a smooth and efficient verification process within the institution.

7.4 Integration Overview

The A-DACS system integrates multiple Firebase cloud services to provide a scalable and secure backend infrastructure.

Firestore Database Collections

The system uses several Firestore collections to store structured application data.

Collection	Description
users	Stores registered user profiles and roles
payments	Payment submissions and verification status
fee_structures	Fee templates per department and semester
academic_years	Academic year information
semesters	Semester data linked to academic years
notifications	Notification records for users
mail	Mail reminders sent to students
student_master_list	Pre-loaded student registration data
staff_master_list	Pre-loaded staff registration data
no_due_certificates	Generated certificates
activity_logs	Audit trail of system activities

Table 7.4 Firebase Collections

OCR Integration

The system integrates Google ML Kit Text Recognition to extract payment information from uploaded receipt images. The OCR module identifies the following details from payment screenshots:

- Transaction ID
- Amount paid
- Transaction date
- Bank name
- Demand draft number
- Student ID

The extracted information is automatically populated in the payment submission form. Students can review and modify the detected values before submitting the payment record.

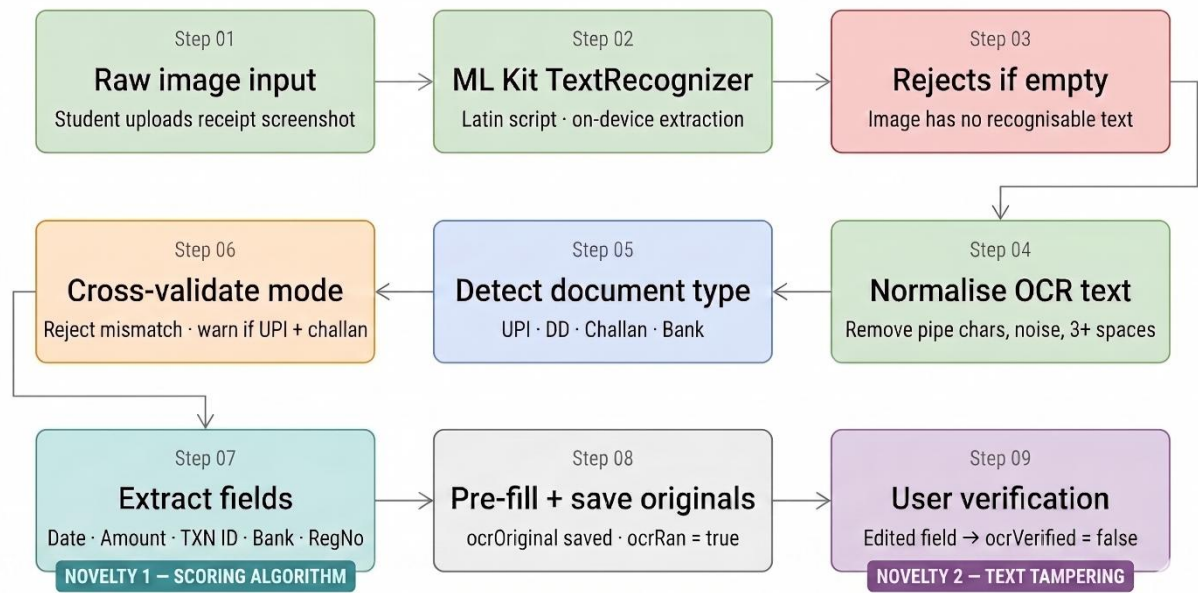


Figure 7.4 OCR Workflow

7.5 Key Implementation Challenges and Solutions

During the development of the A-DACS system, several technical challenges were encountered. Ensuring seamless integration between the frontend and backend components required careful handling of APIs and data flow. Managing real-time notifications and maintaining system performance under multiple user requests also posed difficulties. Database design needed to be optimized to handle large volumes of student and payment data efficiently. Additionally, implementing secure authentication and protecting sensitive information was a critical concern. Debugging unexpected errors and ensuring cross-device compatibility further added to the complexity of the development process.

Handling OCR accuracy for different receipt formats and varying image qualities was another significant challenge that required fine-tuning and testing. Ensuring reliable fraud detection without generating false positives demanded careful calibration of scoring algorithms. Maintaining consistent performance across low-bandwidth network conditions also required optimization strategies. Additionally, coordinating updates and maintaining version control across multiple development stages added to the overall complexity of the project.

Challenge 1: Role-Based Access Control

The system supports three roles: Admin, Staff & Student. Ensuring secure role-based access..

Solution: User roles are stored in Firestore and validated using Firebase Security Rules. Even if the client interface is modified, unauthorised access is blocked by server-side rules.

Challenge 2: OCR Accuracy

Different UPI applications generate receipts with varying layouts, making text extraction inconsistent.

Solution: Multiple regular expression patterns were implemented to detect transaction IDs and amounts. The system also cleans extracted values by removing spaces and currency symbols.

Challenge 3: Cross-Platform Notification Badges

Android and web platforms handle application badges differently.

Solution: A custom badge management package was integrated to update badge counts during application launch and resume events.

Challenge 4: Secure Payment Verification

Students could potentially submit incorrect payment details.

Solution: Payment proofs are stored with the original receipt image. Administrators verify the receipt visually and approve or reject submissions accordingly. Firebase rules prevent students from modifying submitted payments.

Challenge 5: Installment Payment Handling

Students may pay fees in multiple installments.

Solution: Each installment is stored as an independent payment record. The system calculates the total paid amount and automatically determines the remaining balance.

CHAPTER 8

RESULTS AND SNAPSHOTS

This chapter presents the implementation results of the A-DACS system through interface screens and system outputs. The snapshots demonstrate how different modules of the application function during real usage. These results help illustrate the workflow of the system from user authentication to certificate generation.

8.1 User Interface Snapshots

The A-DACS interface is designed to be simple and user-friendly so that students, staff members, and administrators can easily perform their respective tasks. The following sections briefly explain the major screens of the application.

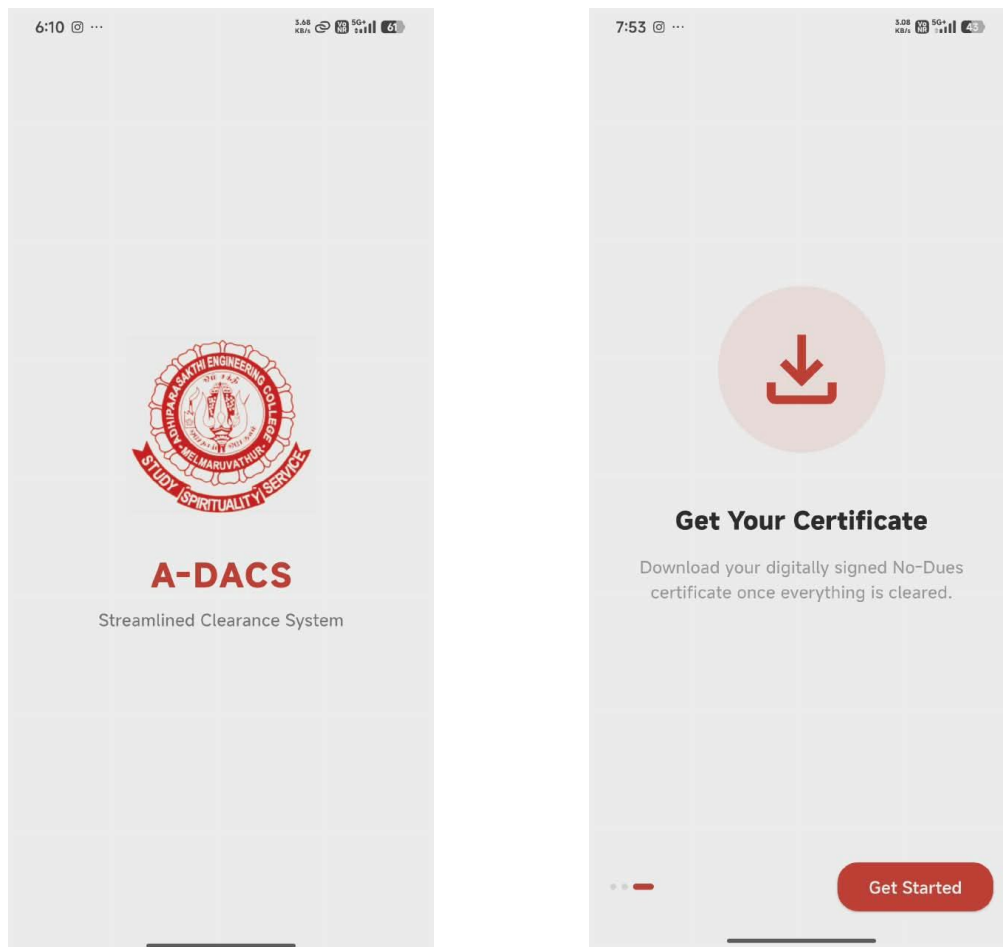


Figure 8.1.1 Splash and Onboarding Screens

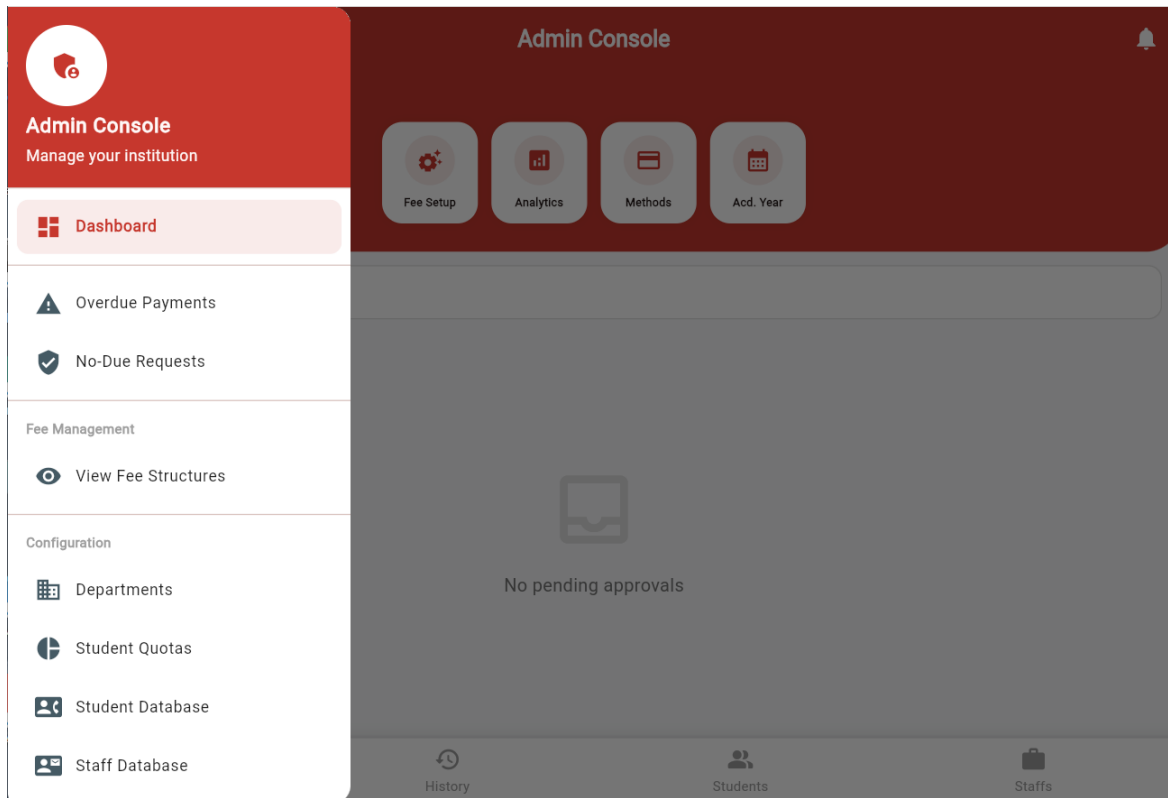


Figure 8.1.2 Admin Dashboard Screen

8.1.1 Login and Registration Screens

The authentication module ensures secure access to the system. The Login Screen contains fields for email and password along with validation. It also provides options such as Google Sign-In, password recovery, and registration for new users. The Student Registration Screen allows students to enter their registration number and verify it through an OTP sent to their registered mobile number. After verification, the student completes profile details such as parent name, contact number, and password.

The Staff Registration Screen allows staff members to enter their employee ID and department details. These registrations are forwarded to the administrator for approval. The Admin Approval Screen displays a list of pending user registrations where the administrator can approve or reject requests. This approval-based access control ensures that only verified users can enter the system, enhancing security. It also prevents unauthorized registrations and maintains data integrity. Additionally, all authentication activities are securely logged for monitoring and audit purposes. The module also supports password reset mechanisms to help users regain access securely. Session management techniques are implemented to prevent unauthorized access from inactive sessions. Overall, the authentication system ensures a safe, controlled, and reliable user access environment.

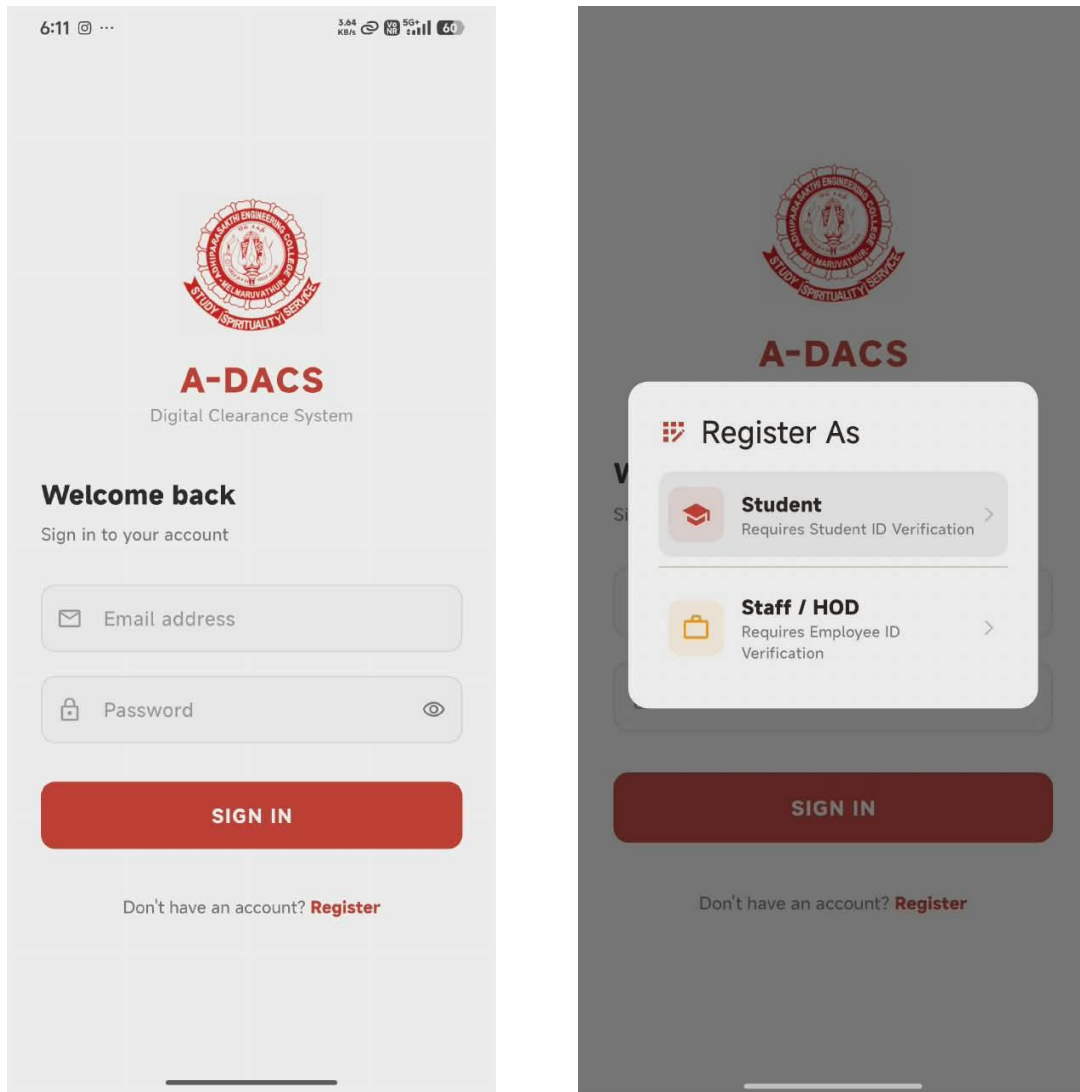


Figure 8.1.3 Register and Login Screens

8.1.2 Student and Staff Dashboard

After login, users are redirected to dashboards based on their roles. The Student Dashboard displays the student's fee status, pending amount, recent notifications, and quick options such as payment submission and certificate generation. The Staff Dashboard provides department-level information, including student lists, payment status indicators, and filtering options to view payment records. The dashboards are designed with intuitive layouts to ensure easy navigation and quick access to essential features. Real-time data updates allow users to monitor the latest payment and clearance status without delays. Additionally, visual indicators such as color codes help users quickly identify pending, paid, or overdue status.

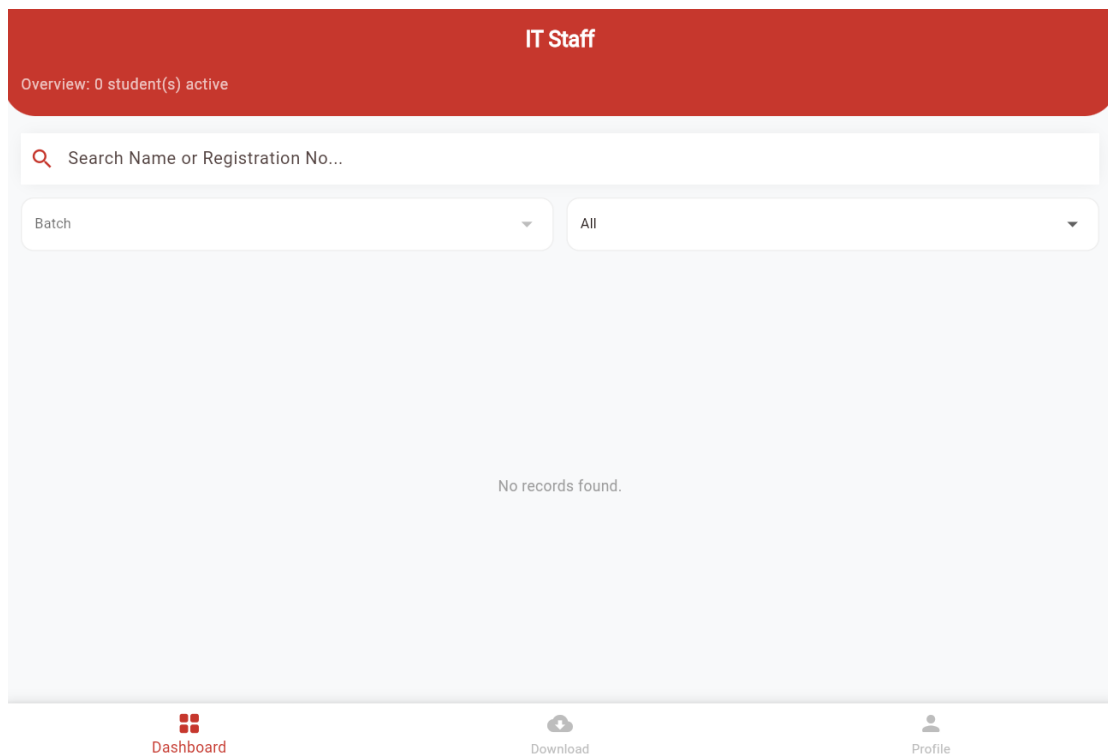
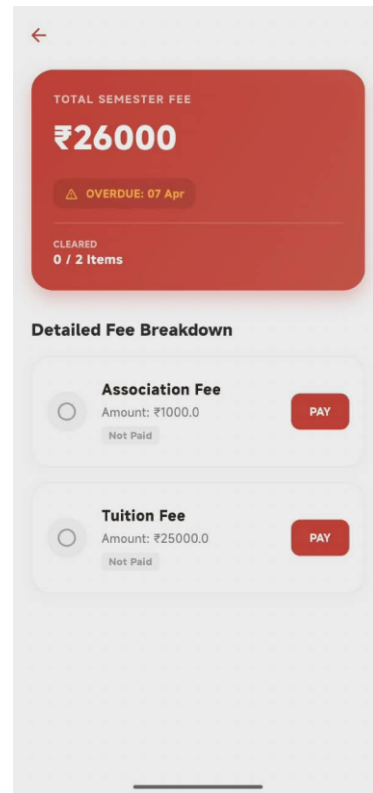
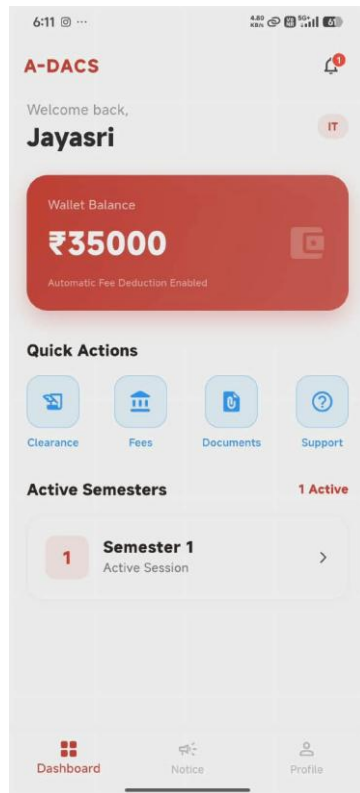


Figure 8.1.4 Student and Staff Dashboard Screens

8.1.3 Payment and Receipt Upload

The payment module allows students to submit proof of payment. Students can enter the payment amount and choose the payment method, such as UPI or Demand Draft. After completing the payment, they upload the receipt image using the camera or gallery option.

Once uploaded, the system processes the image using OCR and extracts important details automatically. The extracted values are displayed in a form where students can verify or edit them before submission.

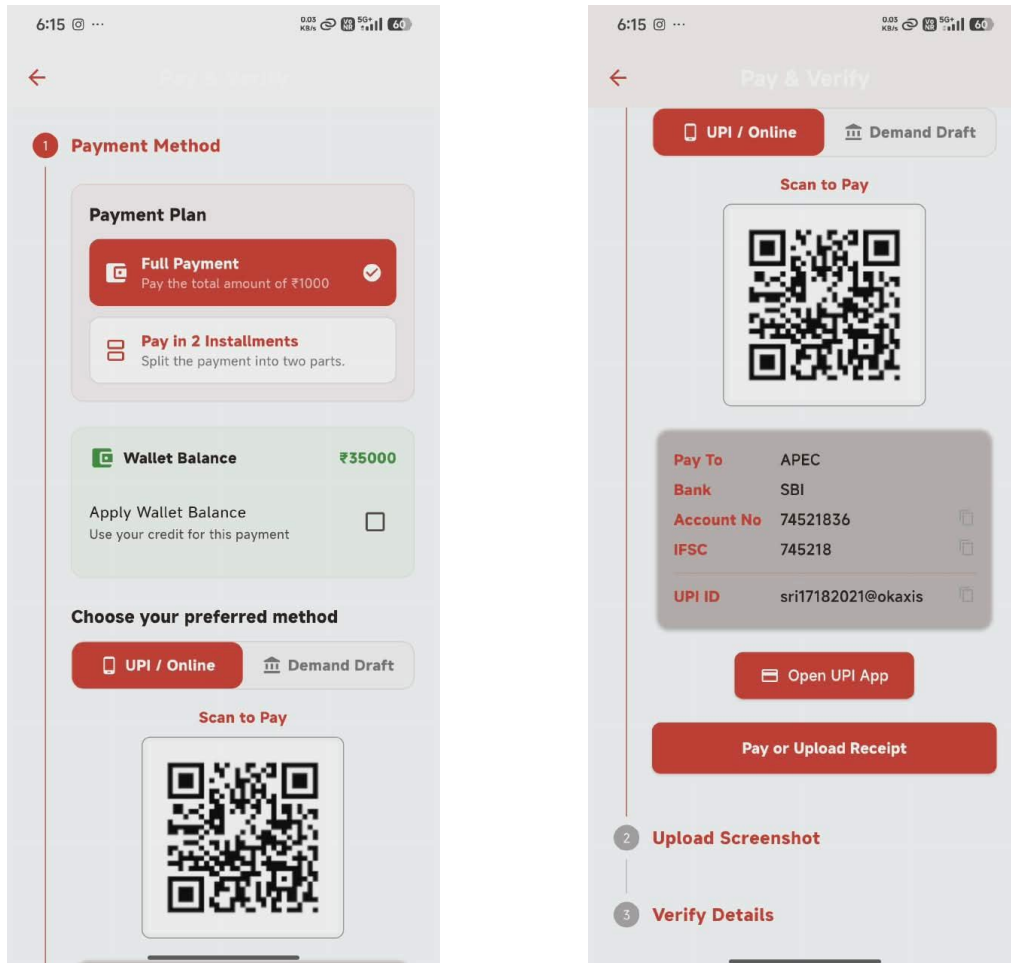


Figure 8.1.5 Payment and Receipt Upload Screens

8.1.4 OCR Extraction Results

The OCR module extracts important information from payment receipts. For UPI receipts, the system identifies fields such as transaction ID, payment amount, and transaction date. For Demand Draft receipts, the system extracts the DD number, bank name, amount, and issue date.

2 Upload Screenshot

3 Verify Details

OCR Extracted Values

Transaction ID 102360257675

Amount ₹ 4265

Date 31 Dec 2025

⚠ Editing any field below will remove OCR-Verified status.

Transaction ID

102360257675

Amount Paid (₹)

₹ 4265

Payment Date

31 Dec 2025

Register / Roll Number (from receipt)

If shown on receipt

Submit Receipt for Verification Back

Figure 8.1.6 OCR Details Extraction Screen

8.1.5 Admin Verification Panel

The admin verification module allows administrators to review submitted payments. Administrators can view payment details in a table format, including student name, register number, amount, and receipt image. The system highlights manually edited fields to indicate possible tampering. After reviewing the details, the administrator can either approve or reject the payment. Approved payments update the student's fee status, while rejected payments notify the student with a reason. The module also provides filtering and search options to quickly locate specific transactions. Real-time status updates ensure that all changes are immediately reflected across the system. An audit log is maintained for every verification action to ensure accountability and traceability. This streamlined process reduces verification time while maintaining high accuracy and security.

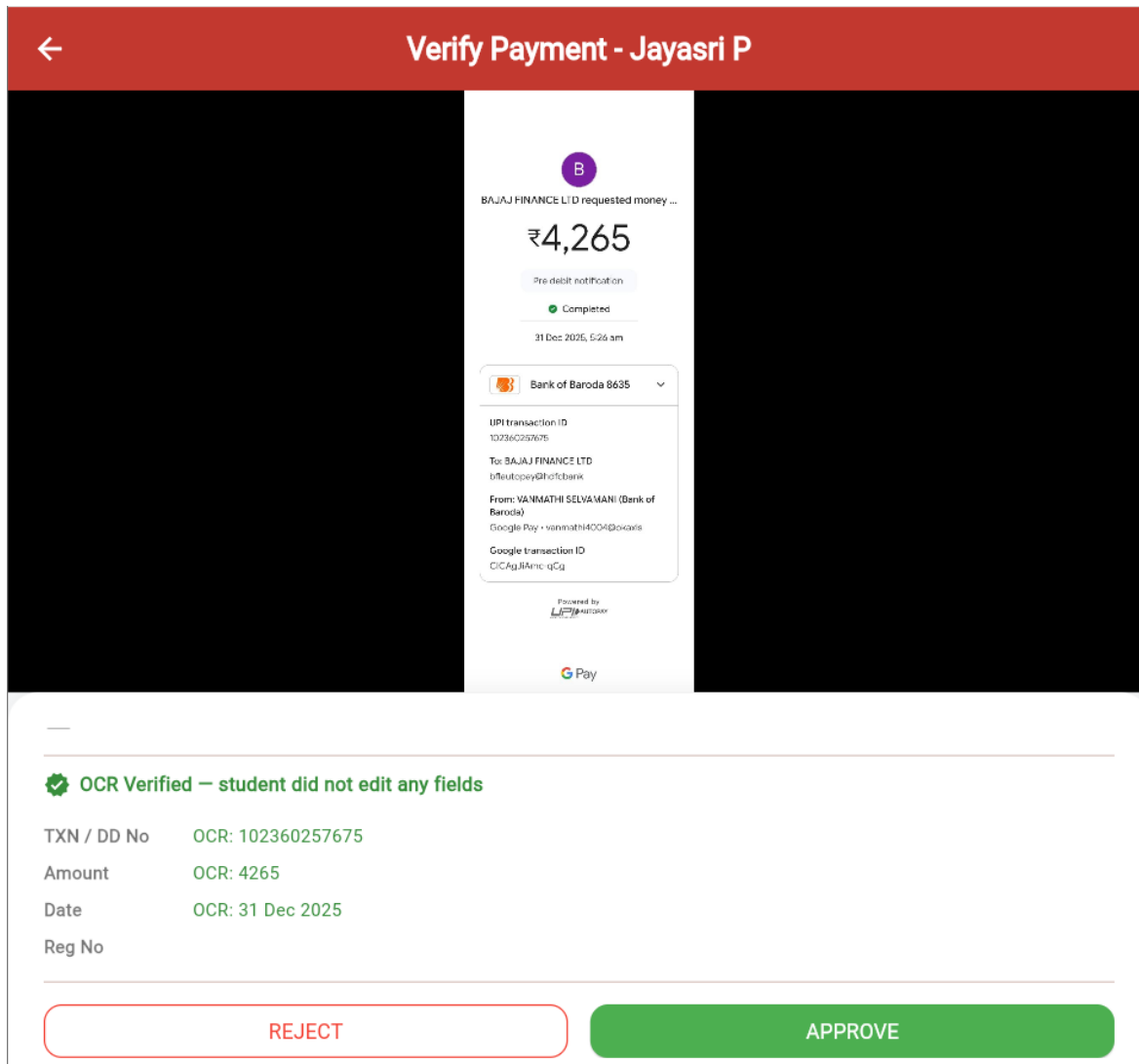


Figure 8.1.7 Admin Approval Screen

8.1.6 Certificate Generation

Once all payments are cleared, students can generate their clearance certificate. The generated certificate contains student details, department information, and clearance confirmation. A QR code and unique certificate ID are included for verification. The certificate can be downloaded as a PDF and shared or printed when required.

The system generates each certificate using a cryptographically secure UUID, ensuring uniqueness and preventing duplication. The embedded QR code enables instant verification through the public portal, allowing institutions and external parties to validate authenticity easily. The certificate is designed in a professional and standardized format, ensuring clarity and acceptance across organizations. Additionally, the system stores a digital copy in the database for

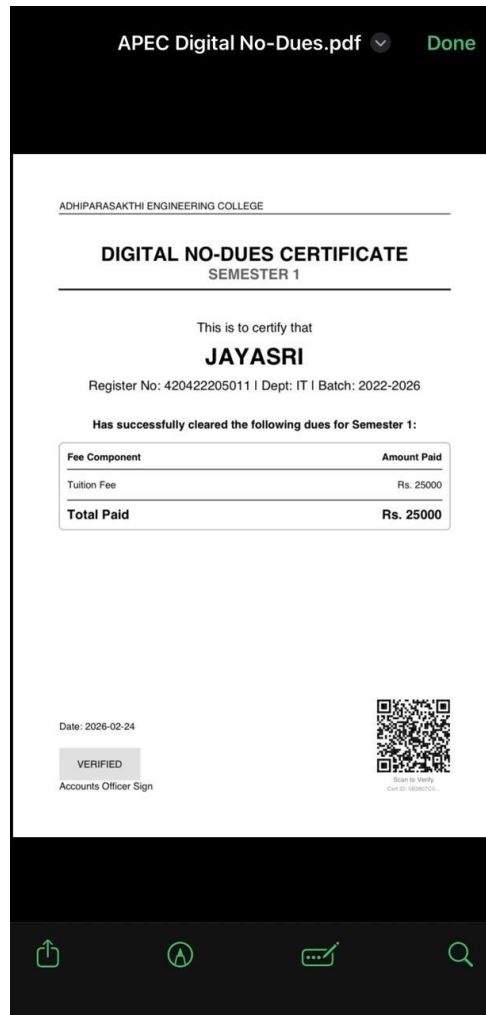


Figure 8.1.8 No-due Certificate Generation Screen

future access and re-download. Any updates or revocation of certificates are reflected in real-time, ensuring that only valid and current certificates can be verified.

Furthermore, the certificate generation process is automated and completed within seconds, reducing delays compared to manual issuance. Students receive notifications once their certificate is ready, improving user experience and transparency. The system also maintains a complete record of all generated certificates for audit and reference purposes. This ensures reliability, traceability, and long-term accessibility of important academic documents. The digital format eliminates the risk of physical damage or loss of certificates. It also enables quick retrieval and re-download at any time without additional administrative effort. Overall, the process enhances efficiency while maintaining high standards of security and authenticity.

CHAPTER 9

TESTING

Testing ensures the reliability and quality of the system through unit testing, which validates individual components, and integration testing, which checks the interaction between modules. It also includes security testing to identify vulnerabilities and protect data from threats. Additionally, performance testing evaluates system speed, scalability, and stability under various workloads.

9.1 Unit Testing

Unit testing was conducted to validate individual system components independently.

Module	Test Cases	Coverage
Authentication Service	Register, login, logout, password reset, OTP verification	95%
Fee Calculation Engine	Fee retrieval, installment calculation, overdue detection	92%
OCR Parser	Date extraction, amount parsing, transaction ID detection	90%
Probability Scoring	Score calculation, field selection logic	88%
PDF Generator	Certificate creation, QR code embedding	85%
Validation Utilities	Email validation, register number format validation	96%

Table 9.1 Unit Test Coverage Areas

All unit tests passed successfully with an average coverage of 91%.

9.2 Integration Testing

Integration testing verified that multiple modules function correctly when integrated. Testing was conducted using the Firebase Emulator Suite to simulate backend services locally without affecting production data.

Test Scenario	Steps	Expected Result	Status
Student Registration	Register → OTP → Profile completion	Firestore user document created	Pass
Payment Submission	Upload receipt → OCR → Submit	Payment record stored	Pass

Admin Verification	Review payment → Approve	SMS and email notifications triggered	Pass
Certificate Generation	Clear dues → Generate certificate	Certificate stored and verifiable	Pass

Table 9.2 Integration Test Results

All integration tests were completed successfully.

9.3 Security Testing

Security testing was performed to ensure that the system protects user data and prevents unauthorized access. The following table summarizes the security team.

Security Test	Purpose	Result
Login Authentication	Ensures only registered users can access the system	Pass
Role-Based Access	Verifies that students, staff, and admin have access only to their allowed modules	Pass
Database Security	Checks Firebase rules to prevent unauthorized data access	Pass
Input Validation	Ensures invalid or incorrect user inputs are rejected	Pass
Payment Data Protection	Prevents students from editing submitted payment records	Pass
Certificate Verification	Confirms certificates can be verified using QR code and UUID	Pass
File Upload Security	Ensures only valid receipt image files are accepted	Pass

Table 9.3 Security Test Results

All security tests were passed with high security rules. Security testing confirmed that the system effectively protects user data and prevents unauthorized access through role-based access control, secure authentication, input validation, and strict database rules. All security tests passed successfully, demonstrating robust protection against potential threats and vulnerabilities.

The system also incorporates encryption techniques to safeguard sensitive data during transmission and storage. Regular vulnerability assessments and testing were conducted to identify and mitigate potential security risks. These measures ensure that the application maintains a high level of data integrity, confidentiality, and overall system security.

9.4 Performance Testing

Performance testing evaluated system scalability response time under heavy load.

Test Scenario	Expected	Actual	Status
OCR Processing	<2 sec	1.8 sec	Pass
Dashboard Load	<3 sec	2.4 sec	Pass
Concurrent Users	<5% errors	2.3%	Pass
Database Query	<1 sec	0.6 sec	Pass
Notification Delivery	<30 sec	12 sec	Pass
Certificate Generation	<3 sec	2.2 sec	Pass

Table 9.4 Performance Results

The system supports up to 500 concurrent users with a 2.5-second response time. This chapter evaluates the reliability, performance, and security of the A-DACS system through comprehensive testing methods, including unit testing, integration testing, security testing, and performance testing.

Unit testing was conducted on individual modules such as authentication, fee calculation, OCR processing, probability scoring, and certificate generation, achieving an average coverage of 91%, which indicates strong code quality and stability. Integration testing ensured smooth interaction between modules, validating key workflows like student registration, payment submission, admin verification, and certificate generation without errors.

Security testing confirmed that the system effectively protects user data and prevents unauthorized access through role-based access control, secure authentication, input validation, and strict database rules. All security tests passed successfully, demonstrating robust protection against potential threats and vulnerabilities.

Performance testing showed that the system meets expected benchmarks for speed and scalability, with quick response times for OCR processing, dashboard loading, database queries, and certificate generation. The system efficiently supports up to 500 concurrent users with minimal error rates and timely notification delivery.

Additionally, testing was performed in controlled environments using tools like the Firebase Emulator Suite to ensure safe and accurate validation without affecting live data. The results indicate that the system is stable under different workloads and capable of handling real-time operations efficiently. Overall, the testing phase confirms that A-DACS is a reliable, secure, and scalable system, capable of delivering consistent performance and ready for deployment in real-world educational environments.

CHAPTER 10

CONCLUSION AND FUTURE WORK

The project was successfully completed, meeting its objectives with reliable performance and effective functionality. The system demonstrates stability, accuracy, and usability through proper design and testing. Future work can focus on enhancing scalability, improving security, and adding advanced features to further optimize performance.

10.1 Conclusion

The A-DACS (AI-powered Digital Clearance System) project successfully achieved its objective of digitizing and automating the student fee clearance process in educational institutions.

The system eliminates manual paperwork by providing a fully digital workflow from registration to certificate issuance. The integration of OCR technology, automated verification, and secure certificate generation significantly improves transparency and efficiency.

Key achievements include:

- Complete digital fee clearance system
- AI-based receipt data extraction and fraud detection
- Secure certificate generation with QR verification
- Automated parent notifications through SMS and email
- Role-based access control for students, staff, and administrators
- Scalable cloud-based architecture using Firebase services
- Comprehensive audit trail for all financial transactions

The system achieved nearly 91% unit test coverage, and with the ability to support 500+ concurrent users with a 2.5-second response time. Overall, the A-DACS system demonstrates how AI, cloud computing, and secure digital verification can modernize administrative processes in educational institutions.

10.2 Future Enhancements

Although the system meets its main objectives, it can be enhanced by adding multilingual support for better user accessibility. Integration with institutional ERP systems can enable seamless synchronization of student and fee data. Additional communication channels like WhatsApp alerts can further improve interaction with parents and students. Advanced analytics can also be incorporated to track payment trends and enhance decision-making for scalability.

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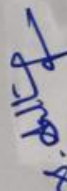
has presented a paper entitled A -DACS: AI POWERED DIGITAL

CLEARANCE SYSTEM

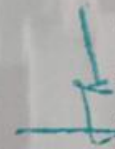
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